

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

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Lesson 22: Electrostatics I

Important Exam Points

Like charges repel each other, while unlike charges attract each other.

Coulomb's Law states that the electrostatic force between two point charges is directly proportional to the product of their charges and inversely proportional to the square of the distance between them.

Coulomb's Law Formula: $F = kq_1q_2/r^2$

The value of Coulomb's constant is $k = 1/4\pi\epsilon_0 = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$.

The permittivity of free space is $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N}\cdot\text{m}^2$.

Charge is measured in **Coulombs (C)**.

Electrostatic force acts along the line joining the two charges.

According to Newton's Third Law, $F_{12} = -F_{21}$.

The resultant force on a charge due to several charges is calculated by the **Superposition Principle**.

Electric charge is **quantized**.

Quantization Formula: $q = ne$

The elementary charge is $e = 1.602 \times 10^{-19} \text{ C}$.

A proton has a charge of $+e$.

An electron has a charge of $-e$.

Charge is always **conserved** in every physical process.

A field is a quantity having a definite value at every point in space and time.

Scalar fields have magnitude only, whereas vector fields have both magnitude and direction.

The electric field is a **vector quantity**.

Electric field is defined as the force acting on a unit positive test charge.

Electric Field Formula: $E = F/q_0$

Electric field due to a point charge: $E = kq/r^2$

Electric field lines originate from positive charges and terminate on negative charges.

The number of electric field lines represents the magnitude of the charge.

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Electric field inside an atom is approximately 10^{11} N/C.

Electric field inside a TV tube is approximately 10^5 N/C.

Electric field in the atmosphere is approximately 10^2 N/C.

Electric field inside a conducting wire is approximately 10^{-2} N/C.

For a particle in equilibrium inside an electric field, $mg = qE$.

Unknown charge Formula: $q = mg/E$

The total electric field due to several charges is the vector sum of individual electric fields.

Electric Field Superposition Formula: $E = E_1 + E_2 + E_3 + \dots$

An electric dipole consists of two equal and opposite charges separated by a small distance.

Dipole Moment Formula: $p = qd$

Electric field on the axial line of a dipole:

$$E = (1/4\pi\epsilon_0)(2p/x^3)$$

For a distant point, the dipole field varies as $1/x^3$.

Torque on an electric dipole is maximum when $\theta = 90^\circ$.

Torque Formula:

$$\tau = pE \sin\theta$$

Torque is zero when the dipole is parallel or antiparallel to the electric field.

20 Difficult Statement, Formula & Numerical MCQs

1. Which statement correctly defines Coulomb's Law?

A) Force is directly proportional to the square of distance.

B) Force depends only on mass.

C) Electrostatic force is directly proportional to the product of charges and inversely proportional to the square of the distance. ✓

D) Force is independent of distance.

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2. If the distance between two point charges becomes three times, the electrostatic force becomes:

- A) Three times
 - B) One-third
 - C) Nine times
 - D) One-ninth ✓
-

3. The SI unit of electric charge is:

- A) Newton
 - B) Joule
 - C) Coulomb ✓
 - D) Volt
-

4. The value of Coulomb's constant is:

- A) 6.67×10^{-11}
 - B) 9.8×10^9
 - C) $8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$ ✓
 - D) 1.602×10^{-19}
-

5. The formula for electric field due to a point charge is:

- A) $E = kr/q$
 - B) $E = qr^2$
 - C) $E = kq/r^2$ ✓
 - D) $E = q/k$
-

6. Which equation represents the quantization of charge?

- A) $q = mv$

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B) $q = IR$

C) $q = ne$ ✓

D) $q = PE$

7. The value of elementary charge is:

A) $8.85 \times 10^{-12} \text{ C}$

B) $1.602 \times 10^{-19} \text{ C}$ ✓

C) $9 \times 10^9 \text{ C}$

D) $6.67 \times 10^{-11} \text{ C}$

8. According to Newton's Third Law in electrostatics:

A) $F_{12} = F_{21}$

B) $F_{12} > F_{21}$

C) $F_{12} = -F_{21}$ ✓

D) $F_{12} = 0$

9. Electric field is defined as:

A) Work done per unit charge

B) Potential per unit charge

C) Force acting on a unit positive test charge ✓

D) Charge per unit force

10. A test charge of 5 C experiences a force of 50 N. The electric field is:

A) 5 N/C

B) 10 N/C ✓

C) 25 N/C

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D) 50 N/C

11. Which statement about electric field lines is correct?

- A) They start from negative charges.
 - B) They form closed loops.
 - C) They start from positive charges and end on negative charges. ✓**
 - D) They have no direction.
-

12. A particle of mass 2 kg is suspended in an electric field of 10 N/C ($g = 10 \text{ m/s}^2$). Its charge is:

- A) 1 C
 - B) 2 C ✓**
 - C) 4 C
 - D) 5 C
-

13. The electric field inside an atom is approximately:

- A) 10^{-2} N/C
 - B) 10^2 N/C
 - C) 10^5 N/C
 - D) 10^{11} N/C ✓**
-

14. Which quantity is a vector?

- A) Charge
 - B) Energy
 - C) Potential
 - D) Electric field ✓**
-

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15. Which principle is used to calculate the net electric field due to many charges?

- A) Pascal's Principle
 - B) Bernoulli's Principle
 - C) Superposition Principle ✓
 - D) Hooke's Law
-

16. The dipole moment is equal to:

- A) q/r
 - B) q^2d
 - C) qd ✓
 - D) qd^2
-

17. The SI unit of dipole moment is:

- A) Joule
 - B) Newton
 - C) Coulomb-meter (C·m) ✓
 - D) Volt-meter
-

18. Torque acting on an electric dipole is given by:

- A) $\tau = pE \cos\theta$
 - B) $\tau = qEd$
 - C) $\tau = pE \sin\theta$ ✓
 - D) $\tau = qE/r$
-

19. Torque on a dipole is maximum when:

- A) $\theta = 0^\circ$

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B) $\theta = 45^\circ$

C) $\theta = 90^\circ$ ✓

D) $\theta = 180^\circ$

20. Which statement about the electric field of a distant dipole is correct?

A) It varies as $1/x$.

B) It varies as $1/x^2$.

C) It varies as $1/x^3$. ✓

D) It remains constant.

Lesson 23: Electrostatics II

Important Exam Points

The electric field due to a continuous charge distribution is obtained by dividing the charge into infinitesimally small elements and integrating their contributions.

The total electric field is the vector sum of all infinitesimal electric fields.

Electric field due to a continuous charge distribution is given by $\mathbf{E} = \int d\mathbf{E}$.

The electric field is a vector quantity and can be resolved into three mutually perpendicular components.

Vector form of electric field: $\mathbf{E} = E_x \hat{i} + E_y \hat{j} + E_z \hat{k}$

For continuous charge distributions, the x, y and z components are calculated separately using integration.

Linear charge density is defined as charge per unit length.

Linear charge density formula: $dq = \lambda ds$

SI unit of linear charge density is C/m.

Surface charge density is defined as charge per unit area.

Surface charge density formula: $dq = \sigma dA$

SI unit of surface charge density is C/m².

Volume charge density is defined as charge per unit volume.

Volume charge density formula: $dq = \rho dV$

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SI unit of volume charge density is C/m^3 .

A uniformly charged ring produces an electric field only along its axis because horizontal components cancel due to symmetry.

Electric field on the axis of a uniformly charged ring is:

$$E = (1/4\pi\epsilon_0) (qz)/(z^2 + R^2)^{3/2}$$

When the observation point is very far from the ring ($z \gg R$), the ring behaves like a point charge.

For $z \gg R$, the electric field becomes:

$$E = (1/4\pi\epsilon_0)(q/z^2)$$

A long uniformly charged wire produces an electric field perpendicular to the wire.

Electric field due to an infinite line charge is:

$$E = \lambda/2\pi\epsilon_0 r$$

The electric field due to an infinite line charge decreases inversely with distance.

Electric flux measures the number of electric field lines passing through a surface.

Electric flux for a uniform electric field is:

$$\Phi = EA$$

If the electric field is not perpendicular to the surface:

$$\Phi = EA \cos\theta$$

General expression of electric flux:

$$\Phi = \int \mathbf{E} \cdot d\mathbf{A}$$

Electric flux is a scalar quantity.

The SI unit of electric flux is $\text{N}\cdot\text{m}^2/\text{C}$.

Flux depends upon both the magnitude of the electric field and the orientation of the surface.

Flux is maximum when the electric field is perpendicular to the surface.

Flux is zero when the electric field is parallel to the surface.

For a spherical surface enclosing a point charge:

$$\Phi = q/\epsilon_0$$

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Gauss's Law states that the total electric flux through any closed surface equals the enclosed charge divided by ϵ_0 .

Gauss's Law Formula:

$$\Phi = \oint \mathbf{E} \cdot d\mathbf{A} = q_{\text{enclosed}}/\epsilon_0$$

Gauss's Law is valid for any closed surface regardless of its shape.

Gauss's Law is equivalent to Coulomb's Law for a point charge.

For a spherical Gaussian surface:

$$\mathbf{E} = (1/4\pi\epsilon_0)(q/r^2)$$

Inside a hollow conducting sphere, the enclosed charge is zero.

The electric field inside a hollow charged sphere is **zero**.

Outside a hollow charged sphere, the electric field behaves as if the entire charge were concentrated at the center.

Gauss's Law greatly simplifies the calculation of electric fields for symmetric charge distributions.

20 Difficult Statement, Formula & Numerical MCQs

1. Which equation represents the electric field due to a continuous charge distribution?

- A) $E = q/r^2$
- B) $E = qV$
- C) $\mathbf{E} = \int d\mathbf{E}$ ✓
- D) $E = Fd$

2. The electric field vector can be resolved into:

- A) Two components only
 - B) Four components
 - C) Three mutually perpendicular components ✓
 - D) Infinite components
-

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3. Which expression represents linear charge density?

- A) $dq = \sigma dA$
 - B) $dq = \rho dV$
 - C) $dq = \lambda ds$ ✓
 - D) $dq = EdA$
-

4. The SI unit of surface charge density is:

- A) C/m
 - B) C/m^2 ✓
 - C) C/m^3
 - D) N/C
-

5. Which expression represents volume charge density?

- A) $dq = \lambda ds$
 - B) $dq = \sigma dA$
 - C) $dq = \rho dV$ ✓
 - D) $dq = q/r^2$
-

6. The electric field on the axis of a uniformly charged ring is:

- A) $E = \lambda/2\pi\epsilon_0 r$
 - B) $E = kq/r^2$
 - C) $E = (1/4\pi\epsilon_0)(qz)/(z^2 + R^2)^{3/2}$ ✓
 - D) $E = pE \sin\theta$
-

7. When $z \gg R$, a charged ring behaves as:

- A) A dipole

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B) An infinite wire

C) A point charge ✓

D) A conductor

8. The electric field due to an infinite line charge is:

A) $\lambda/4\pi\epsilon_0 r^2$

B) $q/4\pi\epsilon_0 r^2$

C) $\lambda/2\pi\epsilon_0 r$ ✓

D) p/x^3

9. Electric flux is defined as:

A) Electric force through a surface

B) Electric potential through a conductor

C) The number of electric field lines passing through a surface ✓

D) Charge per unit area

10. Electric flux for a uniform electric field normal to a surface is:

A) $\Phi = q/\epsilon_0$

B) $\Phi = E/r^2$

C) $\Phi = EA$ ✓

D) $\Phi = qE$

11. The general expression for electric flux is:

A) $\Phi = qE$

B) $\Phi = Er$

C) $\Phi = \int \mathbf{E} \cdot d\mathbf{A}$ ✓

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D) $\Phi = Ed$

12. The SI unit of electric flux is:

A) N/C

B) Volt

C) $\text{N}\cdot\text{m}^2/\text{C}$ ✓

D) Joule

13. Electric flux becomes maximum when:

A) $\theta = 90^\circ$

B) $\theta = 180^\circ$

C) $\theta = 0^\circ$ ✓

D) $\theta = 45^\circ$

14. Which statement correctly defines Gauss's Law?

A) Electric force equals enclosed charge.

B) Electric potential equals electric field.

C) Total electric flux through a closed surface equals enclosed charge divided by ϵ_0 . ✓

D) Electric field depends only on distance.

15. The mathematical form of Gauss's Law is:

A) $E = kq/r^2$

B) $\Phi = EA$

C) $\oint \mathbf{E} \cdot d\mathbf{A} = q_{\text{enclosed}}/\epsilon_0$ ✓

D) $q = ne$

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16. If the enclosed charge inside a Gaussian surface is zero, the net electric flux is:

- A) Infinite
 - B) Positive
 - C) Zero ✓
 - D) Negative
-

17. Inside a hollow charged conducting sphere, the electric field is:

- A) Infinite
 - B) Equal to kq/r^2
 - C) Zero ✓
 - D) Equal to $\lambda/2\pi\epsilon_0 r$
-

18. Outside a hollow charged sphere, the electric field behaves as if:

- A) Charge is uniformly distributed everywhere.
 - B) Charge disappears.
 - C) All charge is concentrated at the center. ✓
 - D) Electric field becomes zero.
-

19. A uniformly charged wire has $\lambda = 8 \times 10^{-6} \text{ C/m}$. Find the electric field at $r = 2 \text{ m}$. ($\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N}\cdot\text{m}^2$)

- A) $3.6 \times 10^4 \text{ N/C}$
 - B) $5.2 \times 10^4 \text{ N/C}$
 - C) $7.2 \times 10^4 \text{ N/C}$ (approximately) ✓
 - D) $9.8 \times 10^4 \text{ N/C}$
-

20. Which statement about Gauss's Law is correct?

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- A) It applies only to spherical surfaces.
- B) It is valid only for point charges.
- C) It is valid for any closed surface regardless of its shape. ✓
- D) It cannot be used for continuous charge distributions.

Lesson 24: Electric Potential Energy

Important Exam Points

Gravitational potential energy is the work done against the gravitational force.

Electrostatic potential energy is the work done against the electrostatic force.

Potential energy is defined only for **conservative forces**.

Change in potential energy is given by $\Delta U = U_b - U_a = -W_{ab}$.

The electrostatic force is a **conservative force**.

Electric potential at infinity is usually taken as **zero**.

Electric potential due to a point charge is:

$$V = (1/4\pi\epsilon_0)(q/r)$$

Electric potential depends only on the distance from the charge.

Electric potential is a **scalar quantity**.

The SI unit of electric potential is **Volt (V)**.

One volt is equal to **1 Joule per Coulomb (1 J/C)**.

Electric potential energy of two point charges is:

$$U = (1/4\pi\epsilon_0)(q_1q_2/r)$$

Positive charges have positive electric potential.

Negative charges have negative electric potential.

The electrostatic force can be either **attractive or repulsive** depending on the signs of the charges.

The force can be obtained from the potential using:

$$\mathbf{F} = -dU/dr$$

Electric field is related to potential by:

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$$E = -dV/dr$$

Positive charges move towards regions of **lower electric potential**.

Negative charges move towards regions of **higher electric potential**.

Only the **potential difference** determines the motion of a charge.

Every physical system tends to minimize its potential energy.

Electric potential exists because the electrostatic force is conservative.

The electric potential at a point due to several charges is the algebraic sum of the individual potentials.

Superposition Formula for Potential:

$$V = V_1 + V_2 + V_3 + \dots$$

General Formula:

$$V = (1/4\pi\epsilon_0) \sum (q_i/r_i)$$

Electric potential is a scalar, so vector addition is **not required**.

The potential due to a dipole is:

$$V = (1/4\pi\epsilon_0)(p \cos\theta/r^2) \text{ (for } r \gg d)$$

The dipole potential becomes zero at:

$$\theta = 90^\circ$$

Dipole moment is given by:

$$p = qd$$

Potential due to a uniformly charged ring is:

$$V = (1/4\pi\epsilon_0)(q/r)$$

One electron volt (eV) is the energy gained by an electron moving through a potential difference of **1 Volt**.

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$$

$$1 \text{ keV} = 10^3 \text{ eV}$$

$$1 \text{ MeV} = 10^6 \text{ eV}$$

$$1 \text{ GeV} = 10^9 \text{ eV}$$

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1 TeV = 10^{12} eV

Electric potential is maximum near positive charges.

Electric potential decreases as the distance from the charge increases.

20 Difficult Statement, Formula & Numerical MCQs

1. Electric potential energy is defined as:

- A) Force acting on a charge.
 - B) Charge per unit work.
 - C) Work done against the electrostatic force. ✓**
 - D) Energy produced by a battery.
-

2. Which equation represents the change in potential energy?

- A) $\Delta U = W$
 - B) $\Delta U = qV$
 - C) $\Delta U = -W_{ab}$ ✓**
 - D) $\Delta U = Fd$
-

3. Electric potential due to a point charge is:

- A) $V = kqr$
 - B) $V = kr^2/q$
 - C) $V = (1/4\pi\epsilon_0)(q/r)$ ✓**
 - D) $V = qr^2$
-

4. The SI unit of electric potential is:

- A) Joule
- B) Coulomb

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C) Volt ✓

D) Newton

5. One Volt is equal to:

A) 1 N/C

B) 1 J/C ✓

C) 1 C/J

D) 1 W/C

6. The electric potential energy of two point charges is:

A) $U = qE$

B) $U = F/q$

C) $U = (1/4\pi\epsilon_0)(q_1q_2/r)$ ✓

D) $U = pE$

7. Electric potential is a:

A) Vector quantity

B) Tensor quantity

C) Scalar quantity ✓

D) Dimensionless quantity

8. Which equation relates force and potential energy?

A) $F = qE$

B) $F = ma$

C) $F = -dU/dr$ ✓

D) $F = qV$

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9. Which equation relates electric field and electric potential?

- A) $E = q/r^2$
- B) $E = qV$
- C) $E = -dV/dr$ ✓
- D) $E = p/r$

10. Positive charges naturally move toward:

- A) Higher potential.
- B) Lower potential. ✓
- C) Equal potential.
- D) Zero potential only.

11. Negative charges naturally move toward:

- A) Lower potential.
- B) Zero potential.
- C) Higher potential. ✓
- D) Constant potential.

12. One electron volt is equal to:

- A) $9 \times 10^9 \text{ J}$
- B) $8.85 \times 10^{-12} \text{ J}$
- C) $1.6 \times 10^{-19} \text{ J}$ ✓
- D) $6.67 \times 10^{-11} \text{ J}$

13. The potential at a point due to many charges is calculated using:

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A) Multiplication Principle.

B) Newton's Law.

C) Superposition Principle. ✓

D) Pascal's Principle.

14. The general formula for electric potential due to several charges is:

A) $V = \Sigma q$

B) $V = qE$

C) $V = (1/4\pi\epsilon_0) \Sigma(q/r_i)$ ✓

D) $V = F/q$

15. The dipole moment is given by:

A) $p = q/r$

B) $p = qr^2$

C) $p = qd$ ✓

D) $p = q^2d$

16. The potential due to a dipole becomes zero at:

A) 0°

B) 45°

C) 90° ✓

D) 180°

17. A charge of $4 \mu\text{C}$ is placed 2 m away from a point. Find the electric potential. ($k = 9 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$)

A) 9000 V

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B) 12000 V

C) 18000 V ✓

D) 36000 V

18. Two charges of $2\ \mu\text{C}$ and $4\ \mu\text{C}$ are separated by 3 m. Their potential energy is approximately:

A) 0.012 J

B) 0.018 J

C) 0.024 J ✓

D) 0.036 J

19. Which statement about electric potential is correct?

A) It is a vector quantity.

B) It depends on direction.

C) It is a scalar quantity and obeys algebraic addition. ✓

D) It has no SI unit.

20. Which statement about electric potential energy is correct?

A) It exists only for non-conservative forces.

B) It is independent of work done.

C) It cannot be negative.

D) It is defined only because the electrostatic force is conservative. ✓

Lesson 25: Capacitors and Currents

Important Exam Points

A capacitor consists of two conductors isolated from each other and their surroundings.

A potential difference between the capacitor plates produces an electric field.

The charge stored on a capacitor is directly proportional to the applied potential difference.

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Capacitance Formula:

$$Q = CV$$

Capacitance is defined as the charge stored per unit potential difference.

Capacitance Formula:

$$C = Q/V$$

The SI unit of capacitance is **Farad (F)**.

One Farad is equal to **1 Coulomb per Volt (1 C/V)**.

The capacitance of a parallel plate capacitor is:

$$C = \epsilon_0 A/d$$

Capacitance is directly proportional to the plate area.

Capacitance is inversely proportional to the distance between the plates.

Increasing the plate area increases the capacitance.

Increasing the distance between the plates decreases the capacitance.

The capacitance depends on the geometry and dimensions of the capacitor.

Capacitors can be connected in parallel or in series.

In a parallel combination, the voltage across each capacitor is the same.

Equivalent capacitance in parallel is:

$$C_{eq} = C_1 + C_2 + \dots + C_n$$

The equivalent capacitance of capacitors in parallel is always greater than the largest individual capacitor.

In a series combination, the charge on each capacitor remains the same.

The total voltage in a series combination is the sum of the individual voltages.

Equivalent capacitance in series is:

$$1/C_{eq} = 1/C_1 + 1/C_2 + \dots + 1/C_n$$

The equivalent capacitance in series is always less than the smallest individual capacitor.

A charged capacitor stores electrical energy.

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Energy stored in a capacitor is:

$$U = \frac{1}{2}CV^2$$

Energy can also be expressed as:

$$U = \frac{1}{2}QV$$

Energy can also be expressed as:

$$U = Q^2/2C$$

The energy stored in a capacitor exists in the electric field between its plates.

Energy density is the energy stored per unit volume.

Energy Density Formula:

$$u = \frac{1}{2}\epsilon_0 E^2$$

Energy density is directly proportional to the square of the electric field.

Wherever an electric field exists, energy is stored.

A dielectric is an insulating material placed between the capacitor plates.

A dielectric becomes polarized in an electric field.

The dielectric weakens the electric field inside the capacitor.

Relative permittivity is represented by ϵ_r .

For air, $\epsilon_r \approx 1.0003$.

For pure water, $\epsilon_r \approx 80$.

The electric field inside a dielectric is:

$$E = (1/4\pi\epsilon_0\epsilon_r)(Q/r^2)$$

The capacitance of a capacitor with dielectric becomes:

$$C = \epsilon_r\epsilon_0 A/d$$

A dielectric increases the capacitance of a capacitor.

The greater the dielectric constant, the greater the capacitance.

20 Difficult Statement, Formula & Numerical MCQs

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1. A capacitor is defined as:

- A) A device that produces electric current.
 - B) A resistor connected to a battery.
 - C) Two conductors separated by an insulating medium. ✓
 - D) A source of magnetic field.
-

2. Which equation defines capacitance?

- A) $C = V/Q$
 - B) $C = Q/V$ ✓
 - C) $C = IV$
 - D) $C = Q^2/V$
-

3. The SI unit of capacitance is:

- A) Volt
 - B) Coulomb
 - C) Farad ✓
 - D) Joule
-

4. The capacitance of a parallel plate capacitor is:

- A) $C = \epsilon_0 d/A$
 - B) $C = A/\epsilon_0 d$
 - C) $C = \epsilon_0 A/d$ ✓
 - D) $C = Qd$
-

5. If the plate separation is doubled, the capacitance becomes:

- A) Double

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

B) Four times

C) Half ✓

D) Unchanged

6. If the plate area is doubled, the capacitance becomes:

A) Half

B) Double ✓

C) One-fourth

D) Unchanged

7. Two capacitors are connected in parallel. Their equivalent capacitance is:

A) Product divided by sum.

B) $C_1 + C_2$ ✓

C) Difference of capacitances.

D) Always zero.

8. Two capacitors are connected in series. The equivalent capacitance is:

A) $C_1 + C_2$

B) $C_1 C_2$

C) $1/C_{eq} = 1/C_1 + 1/C_2$ ✓

D) C_1/C_2

9. In a parallel combination of capacitors, the quantity that remains the same is:

A) Charge

B) Voltage ✓

C) Energy

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

D) Capacitance

10. In a series combination of capacitors, the quantity that remains the same is:

A) Voltage

B) Potential difference

C) Charge ✓

D) Energy density

11. Which expression gives the energy stored in a capacitor?

A) $U = CV$

B) $U = QV$

C) $U = \frac{1}{2}CV^2$ ✓

D) $U = C^2V$

12. The energy stored in a capacitor is present in:

A) The battery.

B) The connecting wires.

C) The electric field between the plates. ✓

D) The dielectric only.

13. The energy density of an electric field is:

A) $u = \epsilon_0 E$

B) $u = \epsilon_0 / E$

C) $u = \frac{1}{2}\epsilon_0 E^2$ ✓

D) $u = QE$

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

14. A dielectric material primarily:

- A) Decreases stored charge.
 - B) Produces electric current.
 - C) Increases the capacitance. ✓
 - D) Removes electric field completely.
-

15. The relative permittivity of pure water is approximately:

- A) 1
 - B) 8
 - C) 80 ✓
 - D) 800
-

16. The relative permittivity of air is approximately:

- A) 0
 - B) 1.0003 ✓
 - C) 10
 - D) 80
-

17. A capacitor has capacitance $4\ \mu\text{F}$ and is connected across $100\ \text{V}$. The stored charge is:

- A) $100\ \mu\text{C}$
 - B) $200\ \mu\text{C}$
 - C) $400\ \mu\text{C}$ ✓
 - D) $800\ \mu\text{C}$
-

18. A capacitor of $5\ \mu\text{F}$ is charged to $20\ \text{V}$. The energy stored is:

- A) $0.0005\ \text{J}$

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

B) 0.001 J ✓

C) 0.002 J

D) 0.004 J

19. If a dielectric of relative permittivity 5 completely fills a capacitor, the capacitance becomes:

A) One-fifth

B) Twice

C) Five times ✓

D) Unchanged

20. Which statement about capacitors is correct?

A) Capacitance depends only on voltage.

B) Capacitance decreases with dielectric insertion.

C) Capacitance depends on plate area, separation, geometry, and dielectric material. ✓

D) Capacitance is independent of plate dimensions.

Lesson 26: Electric Current and Circuits

Important Exam Points

Electric current is the rate of flow of electric charge.

Current Formula:

$$I = dq/dt$$

For constant current, the charge flowing is:

$$q = It$$

The SI unit of current is **Ampere (A)**.

One ampere is equal to **1 Coulomb per second**.

$$1 \text{ A} = 1 \text{ C/s}$$

The conventional direction of current is the direction of positive charge flow.

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

In metallic conductors, electrons move opposite to the direction of conventional current.

Electromotive force (EMF) is the potential difference supplied by a source.

EMF is measured in **Volts (V)**.

Ohm's Law states that current is directly proportional to voltage when temperature remains constant.

Ohm's Law Formula:

$$V = IR$$

A conductor obeys Ohm's Law only if its V-I graph is a straight line.

A curved V-I graph indicates that Ohm's Law is not obeyed.

Resistance is measured in **Ohms (Ω)**.

Current is conserved at every junction in an electric circuit.

Kirchhoff's Current Law (KCL):

$$I_1 = I_2 + I_3$$

In a series combination of resistors, the current through each resistor is the same.

Equivalent resistance in series is:

$$R_{eq} = R_1 + R_2 + \dots + R_n$$

In a parallel combination of resistors, the voltage across each resistor is the same.

Equivalent resistance in parallel is:

$$1/R_{eq} = 1/R_1 + 1/R_2 + \dots + 1/R_n$$

The equivalent resistance in series is greater than any individual resistance.

The equivalent resistance in parallel is less than the smallest individual resistance.

Electrical power is the rate of doing electrical work.

Power Formula:

$$P = dW/dt$$

Power can be expressed as:

$$P = IV$$

Power can also be expressed as:

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$$P = I^2R$$

Power can also be expressed as:

$$P = V^2/R$$

The SI unit of power is **Watt (W)**.

$$1 \text{ Watt} = 1 \text{ Joule/second}$$

Kirchhoff's Voltage Law states that the algebraic sum of potential differences around any closed loop is zero.

Kirchhoff's Voltage Law is based on the conservation of energy.

The time constant of an RC circuit is:

$$\tau = RC$$

Current in a discharging RC circuit is:

$$I = I_0 e^{(-t/RC)}$$

Current decreases exponentially with time in an RC circuit.

The time constant is the time required for the current to fall to $1/e$ ($\approx 36.8\%$) of its initial value.

Drift velocity is the average velocity of charge carriers in a conductor.

Current due to drift velocity is:

$$I = nAev_d$$

Drift velocity Formula:

$$v_d = I/nAe$$

Current density is defined as current per unit cross-sectional area.

Current Density Formula:

$$J = I/A$$

Current density is also expressed as:

$$J = nev_d$$

20 Difficult Statement, Formula & Numerical MCQs

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

1. Electric current is defined as:

- A) Electric force per unit charge.
 - B) Charge stored in a conductor.
 - C) Rate of flow of electric charge. ✓
 - D) Energy supplied per second.
-

2. The SI unit of electric current is:

- A) Volt
 - B) Coulomb
 - C) Ampere ✓
 - D) Watt
-

3. One ampere is equal to:

- A) 1 Joule/second
 - B) 1 Volt/Coulomb
 - C) 1 Coulomb/second ✓
 - D) 1 Newton/Coulomb
-

4. The mathematical expression for current is:

- A) $I = V/R$
 - B) $I = qV$
 - C) $I = dq/dt$ ✓
 - D) $I = P/V$
-

5. Ohm's Law is expressed as:

- A) $P = IV$

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

B) $V = P/R$

C) $V = IR$ ✓

D) $R = VI$

6. Ohm's Law is valid when:

A) Resistance changes continuously.

B) The V-I graph is curved.

C) The V-I graph is a straight line. ✓

D) Current is zero.

7. The equivalent resistance of resistors connected in series is:

A) Product divided by sum.

B) $R_1 + R_2 + \dots + R_n$ ✓

C) Reciprocal sum.

D) Always zero.

8. The equivalent resistance of resistors connected in parallel is:

A) $R_1 + R_2$

B) $R_1 R_2$

C) $1/R_{eq} = 1/R_1 + 1/R_2 + \dots$ ✓

D) $R_1 - R_2$

9. Which quantity remains the same in a series circuit?

A) Voltage

B) Power

C) Current ✓

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

D) Resistance

10. Which quantity remains the same in a parallel circuit?

A) Current

B) Charge

C) Voltage ✓

D) Power

11. Which formula represents electrical power?

A) $P = IR$

B) $P = V/R$

C) $P = IV$ ✓

D) $P = I/V$

12. Another correct expression for electrical power is:

A) $P = VR^2$

B) $P = I/R$

C) $P = I^2R$ ✓

D) $P = R/I$

13. Kirchhoff's Voltage Law states that:

A) Current entering a junction equals current leaving it.

B) Voltage is constant everywhere.

C) The algebraic sum of potential differences around a closed loop is zero. ✓

D) Resistance never changes.

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

14. The time constant of an RC circuit is:

- A) R/C
 - B) C/R
 - C) RC ✓
 - D) $1/RC$
-

15. The current in a discharging RC circuit is given by:

- A) $I = I_0(1 - e^{-t})$
 - B) $I = I_0 t$
 - C) $I = I_0 e^{-(t/RC)}$ ✓
 - D) $I = RC$
-

16. Drift velocity is defined as:

- A) Instantaneous velocity of electrons.
 - B) Maximum speed of electrons.
 - C) Average velocity of charge carriers in a conductor. ✓
 - D) Speed of light inside a wire.
-

17. The current due to drift velocity is:

- A) $I = V/R$
 - B) $I = q/t$
 - C) $I = nAev_d$ ✓
 - D) $I = P/V$
-

18. The current density is given by:

- A) $J = VA$

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B) $J = IR$

C) $J = I/A$ ✓

D) $J = qV$

19. A current of 5 A flows through a resistor of $4\ \Omega$. The power dissipated is:

A) 20 W

B) 50 W

C) 100 W ✓

D) 200 W

20. A wire carries a current of 4 A and has a cross-sectional area of $2 \times 10^{-6}\ \text{m}^2$. The current density is:

A) $5 \times 10^5\ \text{A/m}^2$

B) $1 \times 10^6\ \text{A/m}^2$

C) $2 \times 10^6\ \text{A/m}^2$ ✓

D) $8 \times 10^6\ \text{A/m}^2$

Lesson 27: The Magnetic Field

Important Exam Points

A magnetic field exerts force only on a moving electric charge.

Magnetic force is directly proportional to charge, velocity, magnetic field strength, and $\sin\theta$.

Magnetic Force Formula: $F = qvB \sin\theta$

The magnetic force is always perpendicular to both the velocity and the magnetic field.

Magnetic force is zero when the velocity is parallel to the magnetic field ($\theta = 0^\circ$).

Magnetic force is maximum when the velocity is perpendicular to the magnetic field ($\theta = 90^\circ$).

The SI unit of magnetic field is **Tesla (T)**.

1 Tesla = $1\ \text{N}/(\text{A}\cdot\text{m}) = 10^4\ \text{Gauss}$

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

The Earth's magnetic field is much weaker than one tesla.

When electric and magnetic fields act together, the total force is called the **Lorentz Force**.

Lorentz Force Formula: $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$

The electric force acts even when the charge is at rest, but the magnetic force acts only on moving charges.

A velocity selector allows only particles with a specific speed to pass undeflected.

Velocity Selector Formula: $\mathbf{v} = \mathbf{E}/\mathbf{B}$

A magnetic field can change the direction of motion but cannot change the kinetic energy of a charged particle.

The work done by a magnetic field on a charged particle is always **zero**.

Work Done Formula: $\mathbf{W} = \mathbf{F} \cdot \mathbf{dr} = 0$

A charged particle moving perpendicular to a magnetic field follows a circular path.

Magnetic Force provides the required centripetal force.

Circular Motion Formula: $q\mathbf{vB} = m\mathbf{v}^2/r$

Radius of Circular Path Formula: $\mathbf{r} = m\mathbf{v}/q\mathbf{B}$

The radius increases with increasing mass or velocity.

The radius decreases with increasing magnetic field or charge.

Angular Frequency Formula: $\omega = q\mathbf{B}/m$

A stronger magnetic field produces a smaller circular path.

Earth's magnetic field protects life by deflecting charged particles from the solar wind.

Charged particles are trapped in the **Van Allen Radiation Belts**.

A mass spectrometer separates ions according to their mass-to-charge ratio.

Heavier ions move in larger circular paths than lighter ions.

A current-carrying wire experiences a force inside a magnetic field.

Force on Current-Carrying Wire Formula: $\mathbf{F} = \mathbf{IL} \times \mathbf{B}$

The direction of force on a current-carrying conductor is perpendicular to both current and magnetic field.

Current flowing in a closed loop produces a magnetic field.

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Magnetic Moment Formula: $\mu = IA$

The direction of magnetic moment is perpendicular to the plane of the current loop.

Ampere's Law relates the magnetic field around a closed path to the enclosed current.

Ampere's Law Formula:

$$\oint \mathbf{B} \cdot d\mathbf{s} = \mu_0 I(\text{enclosed})$$

Only the current enclosed inside the loop contributes to Ampere's Law.

Magnetic fields are always produced by moving charges or electric currents.

20 Difficult Statement, Formula & Numerical MCQs

1. The magnetic force acting on a moving charge is given by:

- A) $F = qE$
- B) $F = qv^2B$
- C) $F = qvB \sin\theta$ ✓
- D) $F = qB^2$

2. Magnetic force becomes maximum when:

- A) $\theta = 0^\circ$
- B) $\theta = 30^\circ$
- C) $\theta = 45^\circ$
- D) $\theta = 90^\circ$ ✓

3. The SI unit of magnetic field is:

- A) Weber
- B) Volt
- C) Tesla ✓
- D) Henry

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

4. One tesla is equal to:

- A) 1 J/C
- B) 1 N/C
- C) 1 N/(A·m) ✓
- D) 1 Wb/m²

5. The Lorentz Force is expressed as:

- A) $F = qE$
- B) $F = qvB$
- C) $F = q(E + v \times B)$ ✓
- D) $F = BIL$

6. In a velocity selector, the particle passes straight when:

- A) $v = B/E$
- B) $v = E/B$ ✓
- C) $v = EB$
- D) $v = E + B$

7. A magnetic field can change the:

- A) Mass of a particle
- B) Charge of a particle
- C) Direction of motion only ✓
- D) Kinetic energy

8. The work done by a magnetic field on a charged particle is:

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

- A) Maximum
 - B) Equal to kinetic energy
 - C) Zero ✓**
 - D) Infinite
-

9. A proton enters a magnetic field perpendicular to it. It moves in:

- A) Straight line
 - B) Ellipse
 - C) Circle ✓**
 - D) Parabola
-

10. The radius of the circular path is:

- A) $r = qB/mv$
 - B) $r = qv/mB$
 - C) $r = mv/qB$ ✓**
 - D) $r = q/mvB$
-

11. If the magnetic field doubles while all other quantities remain constant, the radius becomes:

- A) Double
 - B) Half ✓**
 - C) Four times
 - D) Unchanged
-

12. Angular frequency of a charged particle moving in a magnetic field is:

- A) $\omega = mv/q$
- B) $\omega = qv/B$

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

C) $\omega = qB/m$ ✓

D) $\omega = B/q$

13. Which region traps charged particles around Earth?

A) Ozone Layer

B) Ionosphere

C) Van Allen Radiation Belt ✓

D) Troposphere

14. A mass spectrometer separates ions according to their:

A) Color

B) Charge only

C) Mass-to-charge ratio ✓

D) Temperature

15. The force on a current-carrying conductor is given by:

A) $F = qvB$

B) $F = IR$

C) $F = IL \times B$ ✓

D) $F = V/L$

16. Magnetic moment of a current loop is:

A) $\mu = I/L$

B) $\mu = BA$

C) $\mu = IA$ ✓

D) $\mu = qA$

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

17. The direction of magnetic moment is:

- A) Along the current
- B) Opposite to the current
- C) Perpendicular to the plane of the loop ✓
- D) Parallel to the magnetic field

18. Ampere's Law is written as:

- A) $\oint \mathbf{E} \cdot d\mathbf{s} = q/\epsilon_0$
- B) $B = \mu_0 I/r$
- C) $\oint \mathbf{B} \cdot d\mathbf{s} = \mu_0 I(\text{enclosed})$ ✓
- D) $F = qvB$

19. According to Ampere's Law, only _____ contributes to the magnetic field around a closed loop.

- A) External current
- B) Battery voltage
- C) Enclosed current ✓
- D) Resistance

20. A proton moves at 2×10^6 m/s perpendicular to a 0.5 T magnetic field. If its charge is 1.6×10^{-19} C, the magnetic force is:

- A) 8×10^{-15} N
- B) 1.6×10^{-13} N
- C) 1.6×10^{-13} N ✓
- D) 3.2×10^{-12} N

Lesson 28: Electromagnetic Induction

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Important Exam Points

Magnetic flux is the number of magnetic field lines passing perpendicularly through a surface.

Magnetic Flux Formula: $\Phi = BA \cos\theta$

For a non-uniform magnetic field, magnetic flux is calculated by integration.

Flux Integral Formula: $\Phi_B = \int \mathbf{B} \cdot d\mathbf{A}$

The SI unit of magnetic flux is **Weber (Wb)**.

1 Weber = 1 Tesla·m²

The net magnetic flux through any closed surface is always zero.

Gauss's Law for Magnetism Formula: $\oint \mathbf{B} \cdot d\mathbf{A} = 0$

Magnetic monopoles do not exist in nature.

Magnetic field lines always form closed loops.

Faraday's Law states that a changing magnetic flux induces an EMF.

Faraday's Law Formula: $\epsilon = -d\Phi_B/dt$

For a coil having N turns:

$$\epsilon = -N(d\Phi_B/dt)$$

The negative sign in Faraday's Law represents **Lenz's Law**.

Changing magnetic flux can be produced by moving the magnet.

Changing magnetic flux can be produced by moving the coil.

Changing magnetic flux can be produced by changing the area of the coil.

Increasing or decreasing magnetic field strength also changes magnetic flux.

A larger rate of change of magnetic flux produces a larger induced EMF.

Average Induced EMF Formula:

$$\epsilon = -\Delta\Phi/\Delta t$$

EMF is the potential difference produced due to changing magnetic flux.

General EMF Formula:

$$\epsilon = \oint \mathbf{E} \cdot d\mathbf{s}$$

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

A moving conductor inside a magnetic field develops motional EMF.

Motional EMF Formula:

$$\epsilon = BDv$$

Current induced in a closed circuit is given by Ohm's Law.

Induced Current Formula:

$$I = \epsilon/R = BDv/R$$

The induced current depends directly on magnetic field, conductor length, and velocity.

Power dissipated in the circuit is:

$$P = I^2R$$

Electrical power is also:

$$P = IV$$

Mechanical power supplied equals electrical power produced.

Force on a moving conductor is:

$$F = IBD$$

Mechanical Power Formula:

$$P = Fv$$

Lenz's Law states that the induced current always opposes the cause producing it.

The induced magnetic field always opposes the change in magnetic flux.

If magnetic flux increases, the induced current produces an opposing magnetic field.

If magnetic flux decreases, the induced current tries to maintain the original magnetic flux.

Electromagnetic induction is the working principle of generators and transformers.

No induced EMF is produced if magnetic flux remains constant.

20 Difficult Statement, Formula & Numerical MCQs

1. Magnetic flux through a surface is given by:

A) $\Phi = B/A$

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

B) $\Phi = BA$

C) $\Phi = BA \cos\theta$ ✓

D) $\Phi = B^2A$

2. The SI unit of magnetic flux is:

A) Tesla

B) Henry

C) Weber ✓

D) Volt

3. One Weber is equal to:

A) Tesla/m²

B) Newton/Coulomb

C) Tesla·m² ✓

D) Joule/Coulomb

4. According to Gauss's Law for Magnetism, the net magnetic flux through any closed surface is:

A) Maximum

B) Infinite

C) Zero ✓

D) Equal to magnetic charge

5. Which statement is correct regarding magnetic monopoles?

A) They exist inside magnets.

B) They are positive charges.

C) They do not exist in nature. ✓

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

D) They produce electric fields only.

6. Faraday's Law of electromagnetic induction is:

A) $\varepsilon = IR$

B) $\varepsilon = qV$

C) $\varepsilon = -d\Phi/dt$ ✓

D) $\varepsilon = BIL$

7. For a coil having N turns, the induced EMF is:

A) $\varepsilon = N\Phi$

B) $\varepsilon = NIR$

C) $\varepsilon = -N(d\Phi/dt)$ ✓

D) $\varepsilon = NB$

8. The negative sign in Faraday's Law represents:

A) Coulomb's Law

B) Ohm's Law

C) Lenz's Law ✓

D) Ampere's Law

9. Which action does NOT change magnetic flux?

A) Moving the magnet

B) Changing coil area

C) Rotating the coil

D) Keeping the magnet and coil completely stationary ✓

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

10. A loop has an initial flux of 0.8 Wb and final flux of 0.2 Wb in 2 s. The magnitude of induced EMF is:

- A) 0.1 V
 - B) 0.3 V ✓
 - C) 0.5 V
 - D) 1.0 V
-

11. The motional EMF developed in a conductor is:

- A) $\varepsilon = IR$
 - B) $\varepsilon = qE$
 - C) $\varepsilon = \mathbf{B}\mathbf{D}\mathbf{v}$ ✓
 - D) $\varepsilon = B^2D$
-

12. A conductor 0.5 m long moves at 4 m/s perpendicular to a 2 T magnetic field. The induced EMF is:

- A) 2 V
 - B) 3 V
 - C) 4 V ✓
 - D) 8 V
-

13. Induced current in a circuit is calculated using:

- A) Newton's Law
 - B) Coulomb's Law
 - C) $I = \varepsilon/R$ ✓
 - D) Ampere's Law
-

14. If $\varepsilon = 12\text{ V}$ and $R = 4\ \Omega$, the induced current is:

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

A) 2 A

B) 3 A ✓

C) 4 A

D) 6 A

15. Electrical power dissipated in a resistor is:

A) $P = qV$

B) $P = FV$

C) $P = I^2R$ ✓

D) $P = qE$

16. Mechanical power supplied to a moving conductor is:

A) $P = mg$

B) $P = IV^2$

C) $P = Fv$ ✓

D) $P = qv$

17. According to Lenz's Law, the induced current always:

A) Increases magnetic flux.

B) Produces no magnetic field.

C) Opposes the change producing it. ✓

D) Flows in the same direction as the applied current.

18. If magnetic flux through a coil remains constant, the induced EMF is:

A) Maximum

B) Equal to magnetic flux

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

C) Zero ✓

D) Infinite

19. Which device works directly on the principle of electromagnetic induction?

A) Electric heater

B) Capacitor

C) Electric generator ✓

D) Ammeter

20. A coil of 100 turns experiences a flux change of 0.05 Wb in 0.5 s. The induced EMF is:

A) 2 V

B) 5 V

C) 10 V ✓

D) 20 V

Lesson 29: Alternating Current (AC)

Important Exam Points

Alternating Current (AC) is the current that continuously changes its direction with time.

The most common AC waveform is **sinusoidal**.

AC Voltage Formula: $\varepsilon = \varepsilon_m \sin \omega t$

The average value of a complete AC cycle is **zero**.

The average value of $\sin^2 \omega t$ over one complete cycle is **1/2**.

The Root Mean Square (RMS) value represents the effective value of AC.

RMS Voltage Formula: $\varepsilon_{rms} = \varepsilon_m / \sqrt{2} = 0.707 \varepsilon_m$

RMS Current Formula: $I_{rms} = I_m / \sqrt{2} = 0.707 I_m$

Angular Frequency Formula: $\omega = 2\pi / T$

AC is produced by rotating a coil in a magnetic field.

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Magnetic Flux in a Rotating Coil Formula: $\Phi = BA \cos \omega t$

Induced AC EMF Formula: $\varepsilon = BA\omega \sin \omega t$

A transformer works only with alternating current (AC).

A transformer consists of **primary** and **secondary** coils wound on a soft iron core.

Transformer EMF Ratio Formula:

$$\varepsilon_p / \varepsilon_s = N_p / N_s$$

If $N_s > N_p$, the transformer is a **step-up transformer**.

If $N_s < N_p$, the transformer is a **step-down transformer**.

In an ideal transformer, input power equals output power.

Power Relation Formula:

$$\varepsilon_p I_p = \varepsilon_s I_s$$

Current Ratio Formula:

$$I_s / I_p = N_p / N_s$$

Magnetic flux through a coil is directly proportional to current.

Inductance Formula:

$$\Phi = LI$$

The SI unit of inductance is **Henry (H)**.

$$1 \text{ Henry} = 1 \text{ Weber/Ampere}$$

Inductance depends only on the geometry of the coil.

Inductance of a Long Solenoid:

$$L = \mu_0 n^2 l A$$

Changing current in an inductor produces a self-induced EMF.

Self-Induced EMF Formula:

$$\varepsilon = -L(dI/dt)$$

Kirchhoff's Equation for an RL Circuit:

$$\varepsilon = IR + L(dI/dt)$$

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Current growth in an RL circuit:

$$I(t) = (\mathcal{E}/R)(1 - e^{-t/\tau})$$

Time Constant Formula:

$$\tau = L/R$$

After one time constant, current reaches approximately **63%** of its maximum value.

Energy stored in an inductor:

$$U_B = \frac{1}{2}LI^2$$

Energy stored in a capacitor:

$$U_E = \frac{1}{2}CV^2$$

Magnetic Energy Density Formula:

$$u = B^2/2\mu_0$$

In an LC circuit, electrical energy continuously converts into magnetic energy and vice versa.

Total Energy in an LC Circuit:

$$U = \frac{1}{2}LI^2 + q^2/2C$$

Angular Frequency of LC Oscillations:

$$\omega = 1/\sqrt{LC}$$

Charge oscillates according to:

$$q = q_m \cos \omega t$$

In an ideal LC circuit, oscillations continue indefinitely because there is no resistance.

20 Difficult Statement, Formula & Numerical MCQs

1. Alternating current is a current that:

- A) Flows in one direction only
- B) Has constant magnitude
- C) Reverses its direction periodically ✓**
- D) Has zero voltage

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

2. The average value of a complete AC cycle is:

- A) Maximum
- B) Half of peak value
- C) Zero ✓
- D) Equal to RMS value

3. The RMS value of AC voltage is:

- A) ϵ_m
- B) $0.5\epsilon_m$
- C) $\epsilon_m/\sqrt{2}$ ✓
- D) $2\epsilon_m$

4. If the peak voltage is 220 V, the RMS voltage is approximately:

- A) 110 V
- B) 155 V
- C) 156 V ✓
- D) 311 V

5. Angular frequency is related to time period by:

- A) $\omega = \pi T$
- B) $\omega = T/2\pi$
- C) $\omega = 2\pi/T$ ✓
- D) $\omega = T^2$

6. AC is generated by:

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

- A) A stationary magnet
 - B) A battery
 - C) A rotating coil in a magnetic field ✓**
 - D) A resistor
-

7. The induced EMF in a rotating coil is:

- A) $\varepsilon = BA$
 - B) $\varepsilon = BA^2$
 - C) $\varepsilon = BA\omega \sin\omega t$ ✓**
 - D) $\varepsilon = B\omega$
-

8. A transformer operates on:

- A) DC only
 - B) Both AC and DC equally
 - C) AC only ✓**
 - D) Static electricity
-

9. A transformer with $N_s > N_p$ is called:

- A) Current transformer
 - B) Step-down transformer
 - C) Step-up transformer ✓**
 - D) Isolation transformer
-

10. Transformer voltage ratio is:

- A) $\varepsilon_p/\varepsilon_s = N_s/N_p$
- B) $\varepsilon_p/\varepsilon_s = N_p/N_s$ ✓**

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

C) $\epsilon_p = \epsilon_s$

D) $\epsilon = IR$

11. In an ideal transformer, which relation is correct?

A) $\epsilon_p = \epsilon_s$

B) $I_p = I_s$

C) $\epsilon_p I_p = \epsilon_s I_s$ ✓

D) $N_p = N_s$

12. The SI unit of inductance is:

A) Tesla

B) Weber

C) Henry ✓

D) Farad

13. Magnetic flux through a coil is related to current by:

A) $\Phi = IR$

B) $\Phi = CV$

C) $\Phi = LI$ ✓

D) $\Phi = qE$

14. The self-induced EMF is given by:

A) $\epsilon = IR$

B) $\epsilon = BA$

C) $\epsilon = -L(dI/dt)$ ✓

D) $\epsilon = qV$

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

15. The time constant of an RL circuit is:

- A) RC
 - B) CR
 - C) L/R ✓
 - D) R/L
-

16. The energy stored in an inductor is:

- A) CV^2
 - B) q^2/C
 - C) $\frac{1}{2}LI^2$ ✓
 - D) B^2/μ_0
-

17. The energy density of a magnetic field is:

- A) B/μ_0
 - B) $\mu_0 B^2$
 - C) $B^2/2\mu_0$ ✓
 - D) BI
-

18. In an ideal LC circuit, total energy is:

- A) Lost as heat
 - B) Constant electrical energy only
 - C) Constant and exchanged between capacitor and inductor ✓
 - D) Zero
-

19. The angular frequency of an LC circuit is:

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

A) $\omega = \sqrt{LC}$

B) $\omega = LC$

C) $\omega = 1/\sqrt{LC}$ ✓

D) $\omega = L/C$

20. A coil has inductance 2 H and carries a current of 4 A. The energy stored is:

A) 4 J

B) 8 J

C) 16 J ✓

D) 32 J

Lesson 30: Electromagnetic Waves

Important Exam Points

Maxwell combined the laws of electricity and magnetism into **Maxwell's Equations**.

Gauss's Law of Electricity:

$$\oint \mathbf{E} \cdot d\mathbf{A} = Q/\epsilon_0$$

Gauss's Law of Magnetism:

$$\oint \mathbf{B} \cdot d\mathbf{A} = 0$$

The net magnetic flux through any closed surface is **zero**.

Faraday's Law of Electromagnetic Induction:

$$\oint \mathbf{E} \cdot d\mathbf{l} = -d\Phi_B/dt$$

Ampere-Maxwell Law:

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0(I + \epsilon_0 d\Phi_E/dt)$$

The **displacement current** is produced by a changing electric field.

Displacement Current Formula:

$$I_d = \epsilon_0(d\Phi_E/dt)$$

The changing electric field produces a magnetic field.

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

The changing magnetic field produces an electric field.

Electromagnetic waves were predicted by **James Clerk Maxwell**.

Electromagnetic waves were experimentally verified by **Heinrich Hertz**.

Electromagnetic waves do **not require a material medium** for propagation.

Electromagnetic waves travel through vacuum.

The speed of electromagnetic waves in vacuum is:

$$c = 3 \times 10^8 \text{ m/s}$$

Wave Speed Formula:

$$c = f\lambda$$

Frequency Formula:

$$f = c/\lambda$$

Angular Frequency Relation:

$$\omega = kc$$

Electric and magnetic fields are mutually perpendicular.

Both electric and magnetic fields are perpendicular to the direction of wave propagation.

Electromagnetic waves are **transverse waves**.

The electric and magnetic fields are **in phase**.

Relationship between electric and magnetic field amplitudes:

$$E_0 = cB_0$$

The wavelength of visible red light is approximately **700 nm**.

Frequency of red light is approximately:

$$4.29 \times 10^{14} \text{ Hz}$$

Microwave ovens typically use electromagnetic waves having wavelength about **6 cm**.

Radio waves have wavelengths of several meters.

X-rays and Gamma rays have very short wavelengths.

Electromagnetic waves are produced by rapidly changing electric currents.

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

A dipole antenna is commonly used to transmit electromagnetic waves.

The intensity of electromagnetic waves varies as:

$$I \propto 1/r^2$$

Radiation pattern of a dipole antenna:

$$I(\theta) \propto \sin^2\theta/r^2$$

Maximum radiation occurs at:

$$\theta = 90^\circ$$

Minimum radiation occurs at:

$$\theta = 0^\circ$$

An antenna receives electromagnetic waves by converting them into electric current.

The electric field of a polarized wave oscillates in one fixed direction.

Unpolarized light has no fixed direction of electric field.

A polarizer converts unpolarized light into polarized light.

Electromagnetic spectrum includes radio waves, microwaves, infrared, visible light, ultraviolet, X-rays and gamma rays.

20 Difficult Statement, Formula & Numerical MCQs

1. Which Maxwell equation represents Gauss's Law of Electricity?

A) $\oint \mathbf{B} \cdot d\mathbf{A} = 0$

B) $\oint \mathbf{E} \cdot d\mathbf{A} = Q/\epsilon_0$ ✓

C) $\oint \mathbf{E} \cdot d\mathbf{l} = -d\Phi_B/dt$

D) $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I$

2. According to Gauss's Law of Magnetism, the net magnetic flux through a closed surface is:

A) Maximum

B) Infinite

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

C) Zero ✓

D) Equal to electric flux

3. Which equation represents Faraday's Law?

A) $E = mc^2$

B) $c = f\lambda$

C) $\oint \mathbf{E} \cdot d\mathbf{l} = -d\Phi_B/dt$ ✓

D) $F = qE$

4. The displacement current is given by:

A) $I_d = V/R$

B) $I_d = q/t$

C) $I_d = \epsilon_0(d\Phi_E/dt)$ ✓

D) $I_d = BA$

5. Electromagnetic waves travel in vacuum with speed:

A) 3×10^6 m/s

B) 3×10^7 m/s

C) 3×10^8 m/s ✓

D) 3×10^9 m/s

6. The wave equation relating speed, frequency and wavelength is:

A) $c = \lambda/f$

B) $f = \lambda/c$

C) $c = f\lambda$ ✓

D) $\lambda = cf$

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

7. The relationship between electric and magnetic field amplitudes is:

- A) $E_0 = B_0/c$
- B) $B_0 = cE_0$
- C) $E_0 = cB_0$ ✓
- D) $E_0 = c^2B_0$

8. Electromagnetic waves are:

- A) Longitudinal waves
- B) Transverse waves ✓
- C) Mechanical waves
- D) Sound waves

9. In an electromagnetic wave, the electric field and magnetic field are:

- A) Parallel
- B) Opposite
- C) Perpendicular to each other ✓
- D) Independent

10. Red light has wavelength 700 nm. Its frequency is approximately:

- A) 4.29×10^{12} Hz
- B) 4.29×10^{13} Hz
- C) 4.29×10^{14} Hz ✓
- D) 4.29×10^{15} Hz

11. Electromagnetic waves were experimentally verified by:

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

- A) Newton
 - B) Einstein
 - C) Heinrich Hertz ✓**
 - D) Bohr
-

12. Electromagnetic waves were theoretically predicted by:

- A) Faraday
 - B) Coulomb
 - C) James Clerk Maxwell ✓**
 - D) Ampere
-

13. The intensity of electromagnetic radiation decreases with distance according to:

- A) $I \propto r$
 - B) $I \propto r^2$
 - C) $I \propto 1/r^2$ ✓**
 - D) $I \propto 1/r$
-

14. Maximum radiation from a dipole antenna occurs at:

- A) $\theta = 0^\circ$
 - B) $\theta = 45^\circ$
 - C) $\theta = 90^\circ$ ✓**
 - D) $\theta = 180^\circ$
-

15. Microwave ovens generally operate using electromagnetic waves of wavelength approximately:

- A) 6 mm

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

B) 6 cm ✓

C) 6 m

D) 60 cm

16. Which device converts unpolarized light into polarized light?

A) Transformer

B) Capacitor

C) Polarizer ✓

D) Solenoid

17. Which statement about polarized light is correct?

A) Electric field vibrates randomly.

B) Magnetic field disappears.

C) Electric field oscillates in one fixed direction. ✓

D) Wave becomes longitudinal.

18. Which electromagnetic waves have the shortest wavelength?

A) Radio waves

B) Microwaves

C) Visible light

D) Gamma rays ✓

19. The angular frequency of an electromagnetic wave is related to wave number by:

A) $\omega = c/k$

B) $\omega = k/c$

C) $\omega = kc$ ✓

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

D) $\omega = c^2k$

20. Electromagnetic waves are produced by:

- A) Stationary charges
- B) Constant electric fields
- C) **Rapidly changing electric currents** ✓
- D) Uniform magnetic fields

Lesson 31: Light

Important Exam Points

Light is an **electromagnetic wave**.

The speed of light in vacuum is:

$$c = 3 \times 10^8 \text{ m/s}$$

The accurate value of the speed of light is:

$$c = 299,792,458.6 \text{ m/s}$$

The speed of light was first successfully measured by **Ole Roemer** using the eclipse of Jupiter's moon **Io**.

Light of different frequencies corresponds to different colors.

Different wavelengths also correspond to different colors.

Wave Equation:

$$c = f\lambda$$

Visible light occupies only a small part of the electromagnetic spectrum.

Visible light has wavelengths approximately between **400 nm and 700 nm**.

White light contains all visible wavelengths.

The sky appears blue due to the greater scattering of blue light by tiny particles in the atmosphere.

A tungsten bulb emits more yellow light than blue light.

According to **Fermat's Principle**, light always travels along the path requiring the **least time**.

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

For reflection from a mirror:

Angle of Incidence = Angle of Reflection

Reflection Law:

$$\theta_i = \theta_r$$

The speed of light changes when it enters a different medium.

Refractive Index Formula:

$$n = c/v$$

The refractive index is the ratio of the speed of light in vacuum to its speed in the medium.

For most materials:

$$n > 1$$

The refractive index of glass is approximately 1.5.

The refractive index depends on the wavelength of light.

This variation of refractive index with wavelength is called **dispersion**.

A prism separates white light into its component colors due to dispersion.

Light bends toward the normal when entering a denser medium.

Light bends away from the normal when entering a rarer medium.

Snell's Law:

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

When light travels from a denser to a rarer medium and the angle of incidence exceeds the critical angle, **Total Internal Reflection** occurs.

Critical Angle Formula:

$$\theta_c = \sin^{-1}(n_2/n_1)$$

Total Internal Reflection occurs only when light travels from a denser medium to a rarer medium.

Optical fibers work on the principle of **Total Internal Reflection**.

Optical fibers can transmit light over long distances with very little loss.

Optical fibers are widely used in communication systems.

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

20 Difficult Statement, Formula & Numerical MCQs

1. Light is classified as:

- A) Mechanical wave
- B) Longitudinal wave
- C) Electromagnetic wave ✓
- D) Sound wave

2. The speed of light in vacuum is:

- A) 3×10^6 m/s
- B) 3×10^7 m/s
- C) 3×10^8 m/s ✓
- D) 3×10^9 m/s

3. The first successful measurement of the speed of light was made by:

- A) Newton
- B) Maxwell
- C) Ole Roemer ✓
- D) Hertz

4. Which relation connects wavelength, frequency and speed of light?

- A) $c = \lambda f$
- B) $\lambda = cf$
- C) $c = f\lambda$ ✓
- D) $f = \lambda/c$

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

5. The wavelength range of visible light is approximately:

- A) 100–300 nm
 - B) 300–500 nm
 - C) 400–700 nm ✓
 - D) 700–1000 nm
-

6. White light is composed of:

- A) Only red light
 - B) Only blue light
 - C) All visible wavelengths ✓
 - D) Infrared light only
-

7. The sky appears blue mainly because:

- A) Blue light is absorbed.
 - B) Red light is reflected.
 - C) Blue light is scattered more strongly by the atmosphere. ✓
 - D) Green light disappears.
-

8. According to Fermat's Principle, light follows the path of:

- A) Maximum distance
 - B) Minimum distance
 - C) Least time ✓
 - D) Constant velocity
-

9. The law of reflection states that:

- A) $\theta_i > \theta_r$

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

B) $\theta_i < \theta_r$

C) $\theta_i = \theta_r$ ✓

D) $\theta_i = 90^\circ$

10. The refractive index of a medium is defined as:

A) $n = v/c$

B) $n = c/v$ ✓

C) $n = cv$

D) $n = c + v$

11. The approximate refractive index of glass is:

A) 1.0

B) 1.2

C) 1.5 ✓

D) 2.5

12. The separation of white light into different colors by a prism is called:

A) Reflection

B) Diffraction

C) Dispersion ✓

D) Polarization

13. Snell's Law is written as:

A) $n_1\theta_1 = n_2\theta_2$

B) $n_1\cos\theta_1 = n_2\cos\theta_2$

C) $n_1\sin\theta_1 = n_2\sin\theta_2$ ✓

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

D) $\theta_1 = \theta_2$

14. Light bends toward the normal when it enters:

A) A vacuum

B) A rarer medium

C) A denser medium ✓

D) Space

15. Total Internal Reflection occurs when light travels:

A) From air to glass

B) From a denser medium to a rarer medium ✓

C) From vacuum to glass

D) From air to water only

16. The critical angle is calculated using:

A) $\theta_c = \cos^{-1}(n_2/n_1)$

B) $\theta_c = \tan^{-1}(n_2/n_1)$

C) $\theta_c = \sin^{-1}(n_2/n_1)$ ✓

D) $\theta_c = n_1/n_2$

17. Optical fibers operate on the principle of:

A) Diffraction

B) Interference

C) Total Internal Reflection ✓

D) Refraction only

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

18. If the speed of light in a material decreases, its refractive index:

- A) Decreases
- B) Becomes zero
- C) Increases ✓
- D) Remains unchanged

19. A medium has refractive index 1.5. If the speed of light in vacuum is 3×10^8 m/s, the speed of light in the medium is:

- A) 1×10^8 m/s
- B) 2×10^8 m/s ✓
- C) 3×10^8 m/s
- D) 4.5×10^8 m/s

20. Which statement about optical fibers is correct?

- A) They transmit electricity.
- B) They work using reflection from mirrors.
- C) They carry light with very little loss using Total Internal Reflection. ✓

Lesson 32: Interaction of Light with Matter

Important Exam Points

Light interacts with matter in **four basic ways**:

Emission, Absorption, Transmission, and Reflection.

Emission is the process in which matter releases energy as light.

Absorption is the process in which matter absorbs energy from light.

Transmission is the passage of light through a material.

Reflection is the bouncing back of light from a surface.

Hot objects emit **blackbody radiation**.

Blackbody radiation contains **all wavelengths** (continuous spectrum).

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

At temperatures below **800°C**, objects mainly emit **infrared (IR)** radiation.

At about **800°C**, objects appear **red hot**.

At about **2500°C**, objects appear **yellowish-white**.

The wavelength of maximum intensity is given by **Wien's Law**:

$$\lambda_{\max} T = 2.90 \times 10^{-3} \text{ m} \cdot \text{K}$$

As temperature increases, the peak wavelength **decreases**.

Temperature in Kelvin is calculated by:

$$\text{K} = ^\circ\text{C} + 273$$

Blackbody radiation is produced because accelerating charged particles emit electromagnetic waves.

The **Sun** and **stars** closely behave like blackbody radiators.

The **Stefan-Boltzmann Law** gives the total power radiated per unit area:

$$P = \sigma T^4$$

Stefan-Boltzmann Constant:

$$\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{K}^{-4}$$

Hotter objects radiate much more energy than cooler objects.

The temperature of a planet depends upon the Sun's temperature, Sun's radius, and its distance from the Sun.

Each chemical element has its own unique **emission spectrum**.

Hydrogen produces **three visible emission lines**.

Emission spectra are used to identify chemical elements.

Heating a material excites atoms and produces emission spectra.

The **absorption spectrum** contains the same wavelengths as the emission spectrum.

Hydrogen absorbs the same wavelengths that it emits.

Absorption spectra help identify gases present in stars and outer space.

Atmospheric gases such as **oxygen, nitrogen, ozone, and water vapor** absorb specific wavelengths of light.

The colors of objects are produced due to **selective absorption** of certain wavelengths.

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Carotene gives orange, red, and yellow colors.

Chlorophyll gives green color to leaves.

Light is **polarized** when its electric field vibrates in one fixed direction.

Natural light is generally **unpolarized**.

A **polarizer** converts unpolarized light into polarized light.

Amplitude after a polarizer:

$$A = A_0 \cos\theta$$

Intensity after a polarizer:

$$I = I_0 \cos^2\theta$$

Materials having different optical properties in different directions are called **birefringent materials**.

Birefringent materials are used in **LCD watches, calculators, and mobile phone displays**.

20 Difficult Statement, Formula & Numerical MCQs

1. Light interacts with matter mainly through:

A) Two processes

B) Three processes

C) Four processes ✓

D) Five processes

2. Which process allows light to pass through a material?

A) Reflection

B) Emission

C) Transmission ✓

D) Absorption

3. Blackbody radiation consists of:

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

- A) One wavelength only
 - B) Only visible light
 - C) A continuous spectrum of all wavelengths ✓**
 - D) Only infrared radiation
-

4. An object becomes red hot at approximately:

- A) 300°C
 - B) 500°C
 - C) 800°C ✓**
 - D) 2500°C
-

5. At about 2500°C, a heated object appears:

- A) Blue
 - B) Red
 - C) Yellowish-white ✓**
 - D) Green
-

6. Wien's Law is expressed as:

- A) $P = \sigma T^4$
 - B) $\lambda_{\max} T = 2.90 \times 10^{-3} \text{ m} \cdot \text{K}$ ✓**
 - C) $E = hf$
 - D) $c = f\lambda$
-

7. According to Wien's Law, increasing temperature causes the peak wavelength to:

- A) Increase
- B) Remain constant

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

C) Decrease ✓

D) Become zero

8. Temperature in Kelvin is obtained by:

A) $K = ^\circ C - 273$

B) $K = ^\circ C + 273$ ✓

C) $K = ^\circ C \times 273$

D) $K = ^\circ C \div 273$

9. The Stefan-Boltzmann Law is:

A) $\lambda T = \text{constant}$

B) $P = \sigma T^4$ ✓

C) $c = f\lambda$

D) $V = IR$

10. The Stefan-Boltzmann constant is approximately:

A) 3×10^8

B) 2.90×10^{-3}

C) $5.67 \times 10^{-8} \text{ Wm}^{-2}\text{K}^{-4}$ ✓

D) 9.8

11. Which celestial bodies approximately behave as blackbody radiators?

A) Planets only

B) Moons only

C) The Sun and stars ✓

D) Comets only

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

12. Each chemical element has its own unique:

- A) Reflection spectrum
- B) Refraction spectrum
- C) Emission spectrum ✓
- D) Polarization spectrum

13. Hydrogen produces how many visible emission lines according to the lecture?

- A) One
- B) Two
- C) Three ✓
- D) Five

14. The absorption spectrum of hydrogen contains:

- A) Different wavelengths than emission
- B) The same wavelengths as its emission spectrum ✓
- C) Only infrared wavelengths
- D) Only ultraviolet wavelengths

15. Which atmospheric gas absorbs specific wavelengths of light?

- A) Oxygen
- B) Nitrogen
- C) Ozone
- D) All of the above ✓

16. The green color of leaves is mainly due to:

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

A) Carotene

B) Chlorophyll ✓

C) Hemoglobin

D) Melanin

17. A polarizer converts:

A) Polarized light into unpolarized light

B) Unpolarized light into polarized light ✓

C) Infrared into visible light

D) Reflection into refraction

18. The intensity of polarized light after passing through a polarizer is given by:

A) $I = I_0 \sin\theta$

B) $I = I_0 \cos\theta$

C) $I = I_0 \cos^2\theta$ ✓

D) $I = I_0 \tan\theta$

19. Materials having different optical properties in different directions are called:

A) Transparent materials

B) Conductors

C) Birefringent materials ✓

D) Insulators

20. Birefringent materials are commonly used in:

A) Electric heaters

B) Loudspeakers

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

C) LCD watches and mobile phone displays ✓

D) Electric motors

Lesson 33: Interference and Diffraction

Important Exam Points

Interference is the **superposition (addition)** of two or more waves.

When two waves meet **in phase**, **constructive interference** occurs.

Constructive interference produces **maximum amplitude**.

When two waves meet **out of phase**, **destructive interference** occurs.

Destructive interference produces **minimum or zero amplitude**.

Interference occurs for all kinds of waves.

To observe a stable interference pattern, the light source must be **coherent**.

Coherent sources have a constant phase difference.

Young's Double Slit Experiment uses a **single light source** to produce coherent waves.

The interference pattern consists of **alternating bright and dark fringes**.

Bright fringes are produced by **constructive interference**.

Dark fringes are produced by **destructive interference**.

For constructive interference in Young's Double Slit Experiment:

$$d \sin \theta = n\lambda$$

where **n = 0, 1, 2, 3...**

For destructive interference:

$$d \sin \theta = (n + \frac{1}{2}) \lambda$$

where **n = 0, 1, 2, 3...**

The wavelength of light can be calculated using:

$$\lambda = (d \sin \theta) / n$$

The angle is calculated by:

$$\theta = \tan^{-1}(y/L)$$

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where:

y = Fringe distance from the center

L = Distance between slit and screen

When light reflects from a medium of **lower refractive index to higher refractive index**, a $180^\circ (\pi)$ phase change occurs.

When light reflects from a **higher refractive index to lower refractive index**, no phase change occurs.

Thin films produce colorful patterns because of **interference**.

For thin film interference:

Destructive Interference:

$$2nd = m\lambda$$

Constructive Interference:

$$2nd = (m + \frac{1}{2}) \lambda$$

where **m = 0, 1, 2...**

Oil films and soap bubbles show colors because different wavelengths interfere differently.

Diffraction is the bending of light around obstacles or through narrow openings.

Diffraction occurs when the size of the opening is comparable to the wavelength of light.

Diffraction is actually a form of **interference**.

For single slit diffraction:

$$W \sin\theta = m\lambda$$

where **m = $\pm 1, \pm 2, \pm 3...$**

The diffraction angle is:

$$\theta = \sin^{-1}(m\lambda/W)$$

Larger wavelength produces greater diffraction.

Larger slit width produces smaller diffraction.

The resolving power of optical instruments is limited by diffraction.

For a circular aperture, the first dark fringe occurs at:

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

$$\theta_{\min} = 1.22 \lambda/D$$

where:

D = Diameter of the aperture

Larger aperture gives **better resolution**.

Microscopes and telescopes use large apertures to improve image quality.

20 Difficult Statement, Formula & Numerical MCQs

1. Constructive interference occurs when two waves are:

- A) Out of phase
 - B) In phase ✓**
 - C) Perpendicular
 - D) Of different frequencies only
-

2. Destructive interference occurs when waves are:

- A) In phase
 - B) Out of phase ✓**
 - C) Parallel
 - D) Stationary
-

3. Bright fringes are produced due to:

- A) Diffraction
 - B) Constructive interference ✓**
 - C) Reflection
 - D) Refraction
-

4. Dark fringes are produced due to:

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

- A) Reflection
 - B) Refraction
 - C) Destructive interference ✓**
 - D) Polarization
-

5. Stable interference patterns require light sources to be:

- A) Monochromatic only
 - B) Bright
 - C) Coherent ✓**
 - D) Polarized
-

6. The condition for constructive interference is:

- A) $d \sin\theta = (n + \frac{1}{2})\lambda$
 - B) $d \sin\theta = n\lambda$ ✓**
 - C) $W \sin\theta = \lambda$
 - D) $\theta = \tan^{-1}(y/L)$
-

7. The condition for destructive interference is:

- A) $d \sin\theta = n\lambda$
 - B) $d \sin\theta = (n + \frac{1}{2})\lambda$ ✓**
 - C) $W \sin\theta = m\lambda$
 - D) $\theta = \sin^{-1}(\lambda/W)$
-

8. The angle in Young's Double Slit Experiment is calculated using:

- A) $\sin^{-1}$
- B) $\tan^{-1}(y/L)$ ✓**

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

C) $\cos^{-1}$

D) $\tan(yL)$

9. The wavelength is calculated by:

A) $\lambda = n/(d \sin\theta)$

B) $\lambda = (d \sin\theta)/n$ ✓

C) $\lambda = d/n$

D) $\lambda = \theta/n$

10. Reflection from a lower refractive index to a higher refractive index causes a phase change of:

A) 0°

B) 90°

C) $180^\circ (\pi)$ ✓

D) 360°

11. Reflection from a higher refractive index to a lower refractive index causes:

A) 180° phase change

B) No phase change ✓

C) 90° phase change

D) Random phase change

12. The condition for destructive interference in a thin film is:

A) $2nd = m\lambda$ ✓

B) $2nd = (m + \frac{1}{2})\lambda$

C) $d \sin\theta = n\lambda$

D) $W \sin\theta = m\lambda$

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

13. The condition for constructive interference in a thin film is:

- A) $2nd = m\lambda$
- B) $2nd = (m + \frac{1}{2})\lambda$ ✓
- C) $W \sin\theta = \lambda$
- D) $d \sin\theta = \lambda$

14. Colors in soap bubbles are mainly due to:

- A) Refraction
- B) Reflection only
- C) Thin film interference ✓
- D) Polarization

15. Diffraction is defined as:

- A) Reflection of light
- B) Refraction of light
- C) Bending of light around obstacles or narrow openings ✓
- D) Polarization of light

16. The condition for single slit diffraction is:

- A) $d \sin\theta = n\lambda$
- B) $W \sin\theta = m\lambda$ ✓
- C) $2nd = m\lambda$
- D) $n = c/v$

17. The diffraction angle is given by:

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

A) $\tan^{-1}(y/L)$

B) $\theta = \sin^{-1}(m\lambda/W)$ ✓

C) $\theta = \cos^{-1}(m/W)$

D) $\theta = \lambda/W$

18. Increasing the slit width causes diffraction to:

A) Increase

B) Decrease ✓

C) Remain unchanged

D) Become zero

19. The first dark fringe for a circular aperture occurs at:

A) $\theta = \lambda/D$

B) $\theta_{\min} = 1.22 \lambda/D$ ✓

C) $\theta = 2\lambda/D$

D) $\theta = \pi/D$

20. A larger telescope aperture provides:

A) Smaller wavelength

B) Lower magnification

C) Better resolution ✓

D) Less diffraction pattern

Lesson 34: The Particle Nature of Light

Important Exam Points

Light shows both **wave nature** and **particle nature**.

The wave nature of light explains **interference** and **diffraction**.

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Light is an **electromagnetic wave** consisting of electric and magnetic fields.

Electromagnetic waves can travel through **vacuum** without any material medium.

The old concept of **aether** as the medium for light was proven incorrect.

Electromagnetic waves carry both **energy** and **momentum**.

Momentum of an electromagnetic wave is given by:

$$p = U/c$$

Where:

p = momentum

U = energy per unit volume

c = speed of light

Greater wave amplitude means greater energy and momentum.

Light exerts a small force called **radiation pressure** when it strikes a surface.

The **Photoelectric Effect** is the emission of electrons from a metal surface when light falls on it.

Photoelectric emission occurs only if the light frequency is greater than the **threshold frequency**.

Below the threshold frequency, **no electrons are emitted**, regardless of light intensity.

Photoelectrons are emitted **instantaneously** when light of sufficient frequency strikes the metal.

Einstein explained the photoelectric effect by proposing that light consists of tiny packets called **photons**.

A single packet of light is called a **photon**.

The energy of a photon is:

$$E = h\nu$$

or

$$E = h\omega$$

Where:

h = Planck's Constant = $6.626 \times 10^{-34} \text{ J}\cdot\text{s}$

ν = frequency

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

ω = angular frequency

$$h = h/2\pi$$

Higher frequency photons carry more energy.

Microwaves, radio waves, visible light, X-rays and gamma rays are all **photons** of different frequencies.

Electrons inside atoms occupy only **discrete energy levels**.

When an electron moves from a higher energy level to a lower energy level, it **emits a photon**.

The emitted photon has energy equal to the difference between the two energy levels.

When an electron absorbs a photon of the correct energy, it moves to a **higher energy level**.

Photon absorption occurs only when the photon energy exactly matches the energy gap.

Lasers produce light by **stimulated emission**.

Laser light consists of a very large number of photons having **the same frequency**.

Laser light is highly **coherent, monochromatic, and directional**.

Optical pumping is used to excite atoms before laser emission occurs.

20 Difficult Statement, Formula & Numerical MCQs

1. The particle nature of light was successfully explained by:

- A) Newton
- B) Maxwell
- C) Einstein ✓
- D) Hertz

2. The wave nature of light explains:

- A) Photoelectric effect
- B) Interference and diffraction ✓
- C) Radioactivity
- D) Gravitation

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

3. Electromagnetic waves can travel through:

- A) Solids only
- B) Liquids only
- C) Vacuum ✓
- D) Air only

4. The energy of a photon is given by:

- A) $E = mc^2$
- B) $E = pv$
- C) $E = h\nu$ ✓
- D) $E = IR$

5. Planck's constant is equal to:

- A) 9.8 m/s^2
- B) $3 \times 10^8 \text{ m/s}$
- C) $6.626 \times 10^{-34} \text{ J}\cdot\text{s}$ ✓
- D) $1.6 \times 10^{-19} \text{ C}$

6. A photon's energy increases when its:

- A) Wavelength increases
- B) Speed decreases
- C) Frequency increases ✓
- D) Amplitude decreases

7. The photoelectric effect occurs only when:

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

A) Light intensity is high

B) Frequency is above the threshold frequency ✓

C) Wavelength is very long

D) The metal is heated

8. Below the threshold frequency, the photoelectric effect:

A) Increases

B) Produces fewer electrons

C) Does not occur ✓

D) Depends only on intensity

9. Photoelectrons are emitted:

A) After a noticeable delay

B) Only after heating

C) Instantaneously ✓

D) Only in vacuum

10. A single quantum of light is called a:

A) Proton

B) Electron

C) Photon ✓

D) Neutron

11. The momentum carried by electromagnetic waves is related to energy by:

A) $p = mc$

B) $p = mv$

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

C) $p = U/c$ ✓

D) $p = c/U$

12. Which electromagnetic radiation has the highest photon energy?

A) Radio waves

B) Microwaves

C) Visible light

D) Gamma rays ✓

13. Electrons inside an atom occupy:

A) Any energy value

B) Continuous energy levels

C) Discrete energy levels ✓

D) Random energy levels

14. A photon is emitted when an electron:

A) Gains energy

B) Falls to a lower energy level ✓

C) Stops moving

D) Leaves the atom

15. A photon is absorbed when an electron:

A) Moves to a lower energy level

B) Moves to a higher energy level ✓

C) Leaves the nucleus

D) Loses all its energy

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

16. The energy of an emitted photon equals:

- A) The total atomic energy
- B) The nucleus energy
- C) The difference between two energy levels ✓**
- D) Zero

17. Laser light is:

- A) Random in frequency
- B) Incoherent
- C) Monochromatic and coherent ✓**
- D) Invisible only

18. Optical pumping is used to:

- A) Cool atoms
- B) Excite atoms to higher energy levels ✓**
- C) Produce electricity
- D) Increase wavelength

19. If the frequency of light doubles, the photon energy becomes:

- A) Half
- B) Double ✓**
- C) Four times
- D) Unchanged

20. Which statement about photons is correct?

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

- A) Photons have mass and electric charge.
- B) Photon energy depends on light intensity only.
- C) Photon energy depends directly on frequency ($E = hf$). ✓
- D) Photons exist only in visible light. ✓

Lesson 35: Geometrical Optics

Important Exam Points

Geometrical Optics treats light as **straight-line rays** when diffraction and interference can be ignored.

Sharp shadows are evidence that light travels in straight lines.

The **Law of Reflection** states:

Angle of Incidence = Angle of Reflection

$$\theta_i = \theta_r$$

For a rough surface, reflection is called **Diffuse Reflection**.

In diffuse reflection, the angle of incidence equals the angle of reflection at every point, but the surface normals differ.

Polishing a surface reduces diffuse reflection.

A **Plane Mirror** forms a **virtual, upright image**.

The image formed by a plane mirror is at the **same distance behind the mirror** as the object is in front.

A **Virtual Image** cannot be formed on a screen.

A **Real Image** can be formed on a screen.

A **Convex Mirror** has its reflecting surface on the outside.

A **Concave Mirror** has its reflecting surface on the inside.

The **Principal Axis** is the line passing through the center of curvature and the mirror.

The **Radius of Curvature** is represented by **R**.

For a **Concave Mirror**, the focal length is:

$$f = R/2$$

A concave mirror converges parallel rays to a **real focus**.

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

For a **Convex Mirror**, the focal length is:

$$f = -R/2$$

A convex mirror diverges parallel rays and forms a **virtual focus** behind the mirror.

A concave mirror can produce **real or virtual images** depending on the object position.

When the object is beyond the center of curvature, a concave mirror forms a **real, inverted, and diminished image**.

When the object is between the focus and the mirror, a concave mirror forms a **virtual, upright, and magnified image**.

A convex mirror always forms a **virtual, upright, and diminished image**.

Convex mirrors are widely used as **rear-view mirrors** because they provide a wider field of view.

A **Convex Lens** is also called a **Converging Lens**.

A convex lens converges parallel rays to a **real focus**.

A **Concave Lens** is also called a **Diverging Lens**.

A concave lens diverges parallel rays and forms a **virtual focus**.

A concave lens always forms a **virtual, upright, and diminished image**.

The **Lens Maker's Formula** is:

$$1/f = (n - 1) (1/R_1 - 1/R_2)$$

Where:

f = focal length

n = refractive index

R₁ = radius of first surface

R₂ = radius of second surface

A plane surface has an **infinite radius of curvature**.

The **Lateral Magnification** is:

$$M = h'/h$$

Where:

h = object height

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

h' = image height

No lens is perfect because all lenses suffer from **aberration**.

Spherical Aberration occurs when rays passing through different parts of a lens do not meet at the same focus.

Using multiple lenses helps reduce spherical aberration.

20 Difficult Statement, Formula & Numerical MCQs

1. Geometrical optics assumes that light travels as:

- A) Particles only
 - B) Circular waves
 - C) **Straight-line rays** ✓
 - D) Sound waves
-

2. The law of reflection states:

- A) $\theta_i > \theta_r$
 - B) $\theta_i < \theta_r$
 - C) **$\theta_i = \theta_r$** ✓
 - D) $\theta_r = 90^\circ$
-

3. Reflection from a rough surface is called:

- A) Total reflection
 - B) Regular reflection
 - C) **Diffuse reflection** ✓
 - D) Internal reflection
-

4. A plane mirror forms a:

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

A) Real, inverted image

B) Virtual, upright image ✓

C) Real, upright image

D) Inverted virtual image

5. The image formed by a plane mirror is located:

A) Twice the object distance

B) At the focus

C) At the same distance behind the mirror as the object is in front ✓

D) At infinity

6. A virtual image:

A) Can be projected onto a screen

B) Cannot be projected onto a screen ✓

C) Is always inverted

D) Exists only in lenses

7. A concave mirror is also known as a:

A) Diverging mirror

B) Converging mirror ✓

C) Plane mirror

D) Cylindrical mirror

8. The focal length of a concave mirror is:

A) $f = R$

B) $f = -R$

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

C) $f = R/2$ ✓

D) $f = 2R$

9. The focal length of a convex mirror is:

A) $f = R$

B) $f = 2R$

C) $f = -R/2$ ✓

D) $f = -R$

10. A convex mirror always forms an image that is:

A) Real and inverted

B) Real and magnified

C) Virtual, upright and diminished ✓

D) Virtual and enlarged

11. Convex mirrors are commonly used as:

A) Magnifying glasses

B) Camera lenses

C) Rear-view mirrors ✓

D) Projectors

12. A convex lens is also called a:

A) Diverging lens

B) Converging lens ✓

C) Cylindrical lens

D) Plano mirror

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

13. A concave lens is also called a:

- A) Converging lens
- B) Diverging lens ✓**
- C) Plane lens
- D) Convex lens

14. A concave lens always forms an image that is:

- A) Real and magnified
- B) Real and inverted
- C) Virtual, upright and diminished ✓**
- D) Virtual and enlarged

15. The Lens Maker's Formula is:

- A) $1/f = n(R_1 + R_2)$
- B) $f = R/2$
- C) $1/f = (n - 1)(1/R_1 - 1/R_2)$ ✓**
- D) $n = c/v$

16. A plane surface has a radius of curvature equal to:

- A) Zero
- B) One
- C) Infinity ✓**
- D) Negative infinity

17. Lateral magnification is defined as:

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

A) $M = h/h'$

B) $M = h'/h$ ✓

C) $M = f/R$

D) $M = R/f$

18. Spherical aberration occurs because:

A) Light travels too slowly.

B) Mirrors absorb light.

C) Rays passing through different parts of a lens focus at different points. ✓

D) The refractive index becomes zero.

19. Which optical device always produces a wider field of view?

A) Concave mirror

B) Convex lens

C) Convex mirror ✓

D) Plane mirror

20. A concave mirror forms a virtual, upright and magnified image when the object is:

A) Beyond the center of curvature

B) At the center of curvature

C) At the focus

D) Between the focus and the mirror ✓

Lesson 36: Thermal Physics I

Important Exam Points

The ancient theory of heat was called **phlogiston theory**.

Phlogiston was believed to be a weightless fluid stored inside objects.

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

The phlogiston theory was rejected because **heat can be created and destroyed**, unlike a material substance.

Temperature is the physical quantity that indicates the degree of hotness or coldness of a body.

When bodies remain in contact for sufficient time, they attain the **same temperature**.

Heat is a form of energy.

Heat always flows from a body at **higher temperature** to a body at **lower temperature**.

Heat is the transfer of internal energy due to a **temperature difference**.

Thermal equilibrium exists when bodies in thermal contact exchange **no net heat**.

Properties that change with temperature are called **thermometric properties**.

Examples of thermometric properties include:

Expansion of materials

Change in gas pressure

Change in electrical resistance

Change in color

A **constant-volume gas thermometer** measures temperature by measuring gas pressure.

The lowest possible temperature is called **absolute zero**.

Absolute zero is:

$$0 \text{ K} = -273.15^\circ\text{C} = -459.67^\circ\text{F} \text{ (approximately } -460^\circ\text{F)}$$

The Kelvin scale is the SI temperature scale used in scientific calculations.

Water freezes at:

$$0^\circ\text{C} = 273.15 \text{ K} = 32^\circ\text{F}$$

Water boils at:

$$100^\circ\text{C} = 373.15 \text{ K} = 212^\circ\text{F}$$

Temperature conversion formulas:

$$^\circ\text{F} = (9/5 \times ^\circ\text{C}) + 32$$

$$^\circ\text{C} = (5/9) (^\circ\text{F} - 32)$$

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

$$K = ^\circ C + 273.15$$

The surface temperature of the Sun is approximately:

$$5500 \text{ K}$$

The temperature of a hydrogen bomb is approximately:

$$10^8 \text{ K}$$

The lowest temperature ever achieved experimentally is about:

$$10^{-7} \text{ K}$$

James Prescott Joule proved that mechanical work can be converted into heat.

One calorie is defined as the heat required to raise the temperature of **1 g of water** by **1°C**.

Conversion:

$$1 \text{ cal} = 4.186 \text{ J}$$

$$1 \text{ kcal} = 1000 \text{ cal}$$

Most materials **expand when heated**.

Linear Expansion Formula:

$$\Delta L = \alpha L_0 \Delta T$$

or

$$L = L_0(1 + \alpha \Delta T)$$

Where α is the coefficient of linear expansion.

Coefficient of Linear Expansion:

$$\alpha = \Delta L / (L_0 \Delta T)$$

Volume Expansion Formula:

$$\Delta V = \beta V_0 \Delta T$$

Coefficient of Volume Expansion:

$$\beta = \Delta V / (V_0 \Delta T)$$

Relationship between linear and volume expansion coefficients:

$$\beta = 3\alpha$$

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Different materials have different expansion coefficients.

A **bimetallic strip** bends when heated because its two metals expand by different amounts.

Bimetallic strips are used in **thermostats**, electric irons, and refrigerators.

Water shows **anomalous expansion** between **0°C and 4°C**.

Water contracts when heated from **0°C to 4°C**.

Water has maximum density at **4°C**.

Heat Capacity Formula:

$$C = Q / \Delta T$$

Specific Heat Formula:

$$Q = mc\Delta T$$

Where:

Q = Heat added

m = Mass

c = Specific heat capacity

ΔT = Temperature change

The specific heat capacity of water is:

$$1 \text{ cal/g}^\circ\text{C} = 4186 \text{ J/kg}^\circ\text{C}$$

Water has a much higher specific heat than most metals.

Materials with low specific heat heat up and cool down more quickly.

Heat is transferred by **three methods**:

Conduction

Convection

Radiation

Conduction transfers heat without movement of matter.

Heat Conduction Formula:

$$Q = kA(\Delta T/L)t$$

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Where:

k = Thermal conductivity

A = Cross-sectional area

ΔT = Temperature difference

L = Length

t = Time

Materials with high thermal conductivity (like metals) conduct heat efficiently.

Cloth is a poor conductor of heat and acts as an insulator.

Convection transfers heat by the movement of fluids.

Radiation transfers heat through electromagnetic waves and requires no medium.

20 Difficult Statement, Formula & Numerical MCQs

1. The ancient theory that considered heat to be a weightless fluid was called:

A) Caloric theory

B) Phlogiston theory ✓

C) Kinetic theory

D) Atomic theory

2. Heat always flows from:

A) Cold body to hot body

B) Hot body to cold body ✓

C) Equal temperatures to unequal temperatures

D) Higher pressure to lower pressure

3. Thermal equilibrium exists when:

A) Both bodies have equal mass

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

B) Both bodies have equal volume

C) No net heat flows between bodies ✓

D) Their pressures are equal

4. The SI unit of temperature is:

A) Celsius

B) Fahrenheit

C) Kelvin ✓

D) Rankine

5. Absolute zero corresponds to:

A) 100°C

B) -100°C

C) -273.15°C ✓

D) 273°C

6. Which temperature scale is used in scientific calculations?

A) Fahrenheit

B) Celsius

C) Kelvin ✓

D) Centigrade only

7. The correct Celsius to Kelvin conversion is:

A) $K = ^\circ C - 273.15$

B) $K = ^\circ C + 273.15$ ✓

C) $K = (9/5)^\circ C$

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

D) $K = (5/9)^{\circ}C$

8. Who demonstrated the mechanical equivalent of heat?

A) Newton

B) Maxwell

C) James Prescott Joule ✓

D) Einstein

9. One calorie is equal to:

A) 2.186 J

B) 3.186 J

C) 4.186 J ✓

D) 5.186 J

10. The formula for linear expansion is:

A) $\Delta L = \beta L_0 \Delta T$

B) $\Delta L = \alpha L_0 \Delta T$ ✓

C) $\Delta L = mc \Delta T$

D) $\Delta L = kA \Delta T$

11. The relationship between β and α is:

A) $\beta = \alpha$

B) $\beta = 2\alpha$

C) $\beta = 3\alpha$ ✓

D) $\beta = \alpha^2$

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

12. Water shows anomalous expansion between:

- A) 4°C and 8°C
 - B) 0°C and 4°C ✓
 - C) 10°C and 20°C
 - D) -4°C and 0°C
-

13. Water has maximum density at:

- A) 0°C
 - B) 2°C
 - C) 4°C ✓
 - D) 100°C
-

14. Heat capacity is defined as:

- A) $C = m\Delta T$
 - B) $C = Q/\Delta T$ ✓
 - C) $C = Qm$
 - D) $C = kA$
-

15. The formula relating heat, mass, and specific heat is:

- A) $Q = mL$
 - B) $Q = kA\Delta T$
 - C) $Q = mc\Delta T$ ✓
 - D) $Q = PV$
-

16. The specific heat capacity of water is approximately:

- A) $1000 \text{ J/kg}^{\circ}\text{C}$

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

B) 2000 J/kg°C

C) 4186 J/kg°C ✓

D) 8400 J/kg°C

17. Heat transfer without movement of matter is called:

A) Convection

B) Radiation

C) Conduction ✓

D) Diffusion

18. Which equation represents heat conduction?

A) $Q = mc\Delta T$

B) $Q = PV$

C) $Q = kA(\Delta T/L)t$ ✓

D) $Q = \sigma T^4$

19. Bimetallic strips work because:

A) Both metals expand equally.

B) One metal melts first.

C) Different metals expand at different rates. ✓

D) They conduct electricity differently.

20. Which method of heat transfer does not require a material medium?

A) Conduction

B) Convection

C) Radiation ✓

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

D) Diffusion

Lesson 37: Thermal Physics II

Important Exam Points

A **system** is the object or substance being studied in thermodynamics.

The condition of a system is described by **thermodynamic variables**.

The main thermodynamic variables are:

Pressure (P)

Volume (V)

Temperature (T)

The relationship among pressure, volume, and temperature is called the **Equation of State (EOS)**.

For an ideal gas:

$$PV = NkBT$$

Where:

P = Pressure

V = Volume

N = Number of molecules

kB = Boltzmann constant

T = Absolute temperature (Kelvin)

Thermodynamics is the study of heat and energy transfer.

The **First Law of Thermodynamics** is based on the law of conservation of energy.

First Law of Thermodynamics:

$$\Delta E = \Delta q + \Delta w$$

Where:

ΔE = Change in internal energy

Δq = Heat added to the system

Δw = Work done on the system

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Heat added to the system is taken as **positive**.

Heat leaving the system is taken as **negative**.

Work done on the system is **positive**.

Work done by the system is **negative**.

Heat and work are **path variables** because they depend on the process followed.

Internal energy is a **state variable** because it depends only on the initial and final states.

The work done by a gas during expansion is:

$$dw = -P_{ext} dV$$

The total work done during expansion is:

$$w = -\int P_{ext} dV$$

If a gas expands into a vacuum:

$$P_{ext} = 0$$

Therefore:

$$w = 0$$

For expansion against constant external pressure:

$$w = -P_{ext}\Delta V$$

During reversible expansion of an ideal gas:

$$w = -NkBT \ln(V_2/V_1)$$

The internal energy of an ideal gas is:

$$E = (3/2)NkBT$$

The internal energy of an ideal gas depends only on:

Temperature

Number of molecules

During free expansion of an ideal gas:

$$\Delta E = 0$$

At constant volume:

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

$$\Delta Q_V = C_V \Delta T$$

Since no expansion occurs:

$$\Delta W = 0$$

Therefore:

$$\Delta Q_V = \Delta E$$

C_V is called the **specific heat at constant volume**.

At constant pressure:

$$\Delta Q_P = C_P \Delta T$$

C_P is called the **specific heat at constant pressure**.

For an ideal gas:

$$C_P = C_V + Nk_B$$

For an ideal gas:

$$C_V = (3/2)Nk_B$$

Therefore:

$$C_P = (5/2)Nk_B$$

The ratio of specific heats is:

$$\gamma = C_P / C_V$$

An **adiabatic process** is one in which:

$$Q = 0$$

For an adiabatic process:

$$TV^{(\gamma-1)} = \text{Constant}$$

The **Second Law of Thermodynamics** states that heat cannot flow naturally from a colder body to a hotter body.

Heat naturally flows from:

Higher temperature $\rightarrow$ Lower temperature

The Second Law proves that a **perpetual motion machine is impossible**.

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

A **thermal reservoir** is a large body whose temperature remains nearly constant during heat exchange.

Examples of thermal reservoirs:

The human body

The atmosphere

Large lakes

Heat engines operate between:

A hot reservoir

A cold reservoir

A heat engine absorbs heat from the hot reservoir and rejects some heat to the cold reservoir.

Work produced by a heat engine is:

$$W = Q_{in} - Q_{out}$$

Efficiency of a heat engine:

$$\eta = W/Q_{in}$$

or

$$\eta = (Q_{in} - Q_{out})/Q_{in}$$

Maximum efficiency of a heat engine:

$$\eta = 1 - T_C/T_H$$

Where:

T_H = Temperature of hot reservoir (Kelvin)

T_C = Temperature of cold reservoir (Kelvin)

The efficiency of any heat engine is always:

Less than 1 (less than 100%)

No heat engine can convert all supplied heat into useful work.

A **refrigerator** is a heat engine operating in reverse.

A refrigerator uses external work to transfer heat from a cold region to a hot region.

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

The Second Law also states that a refrigerator cannot transfer heat from cold to hot without external work.

20 Difficult Statement, Formula & Numerical MCQs

1. The relationship between pressure, volume, and temperature is called:

- A) Boyle's Law
 - B) Charles' Law
 - C) Equation of State (EOS) ✓
 - D) Hooke's Law
-

2. The equation of state for an ideal gas is:

- A) $PV = nRT^2$
 - B) $P = Nk_B$
 - C) $PV = Nk_B T$ ✓
 - D) $PT = Nk_B V$
-

3. The First Law of Thermodynamics is written as:

- A) $\Delta E = \Delta q - \Delta w$
 - B) $\Delta E = \Delta q + \Delta w$ ✓
 - C) $\Delta E = W/Q$
 - D) $\Delta E = PV$
-

4. Heat and work are called:

- A) State variables
- B) Path variables ✓
- C) Intensive variables

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

D) Extensive variables

5. Internal energy is a:

A) Path variable

B) State variable ✓

C) Constant variable

D) Random variable

6. The work done during gas expansion is:

A) $dw = PdV$

B) $dw = -P_{ext} dV$ ✓

C) $dw = VdP$

D) $dw = mc\Delta T$

7. If a gas expands freely into a vacuum, the work done is:

A) Positive

B) Negative

C) Zero ✓

D) Infinite

8. The internal energy of an ideal gas is:

A) $NkBT$

B) $2NkBT$

C) $(3/2)NkBT$ ✓

D) $(5/2)NkBT$

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

9. The internal energy of an ideal gas depends upon:

- A) Pressure only
 - B) Volume only
 - C) Temperature and number of molecules ✓
 - D) Pressure and volume only
-

10. At constant volume, heat supplied is given by:

- A) $Q = CP\Delta T$
 - B) $Q = CV\Delta T$ ✓
 - C) $Q = PV$
 - D) $Q = mL$
-

11. At constant pressure, heat supplied is:

- A) $Q = CV\Delta T$
 - B) $Q = CP\Delta T$ ✓
 - C) $Q = kA\Delta T$
 - D) $Q = PV$
-

12. For an ideal gas:

- A) $CP = CV$
 - B) $CP = CV - Nk_B$
 - C) $CP = CV + Nk_B$ ✓
 - D) $CP = 2CV$
-

13. The ratio γ is defined as:

- A) CV/CP

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

B) C_P/C_V ✓

C) P/V

D) V/P

14. An adiabatic process is one in which:

A) Pressure remains constant.

B) Temperature remains constant.

C) No heat is exchanged ($Q = 0$). ✓

D) Volume remains constant.

15. The adiabatic equation is:

A) $PV = \text{Constant}$

B) $P/T = \text{Constant}$

C) $TV^{(\gamma-1)} = \text{Constant}$ ✓

D) $VT = \text{Constant}$

16. The Second Law of Thermodynamics states that heat naturally flows:

A) From cold to hot

B) From hot to cold ✓

C) In both directions equally

D) Only in gases

17. A thermal reservoir is a body that:

A) Has zero temperature

B) Maintains nearly constant temperature during heat exchange ✓

C) Stores electricity

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

D) Has no heat capacity

18. The work produced by a heat engine is:

A) $W = Q_{in} + Q_{out}$

B) $W = Q_{in} - Q_{out}$ ✓

C) $W = Q_{in}/Q_{out}$

D) $W = Q_{out} - Q_{in}$

19. The maximum efficiency of a heat engine is:

A) $\eta = T_H/T_C$

B) $\eta = T_C/T_H$

C) $\eta = 1 - T_C/T_H$ ✓

D) $\eta = T_H - T_C$

20. A refrigerator works by:

A) Producing heat without work

B) Converting all heat into work

C) Using external work to transfer heat from a cold region to a hot region ✓

Lesson 38: Thermal Physics III

Important Exam Points

Heat is the **random kinetic energy of atoms and molecules**.

A hot gas has atoms moving faster on average than a cold gas.

The study of heat based on atomic and molecular motion is called **Statistical Mechanics**.

Statistical Mechanics explains macroscopic properties using the behavior of individual molecules.

Temperature is related to the **average kinetic energy** of atoms.

Heat is the **energy transferred** from one object to another because of a temperature difference.

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For an ideal gas:

$$PV = Nk_B T$$

Pressure of an ideal gas increases when temperature increases because molecules strike the container walls more frequently and with greater force.

At constant pressure:

$$V \propto T$$

Absolute zero (**0 K**) corresponds to zero molecular motion in the ideal gas model.

Pressure is defined as:

$$P = F/A$$

Newton's Second Law:

$$F = \Delta p / \Delta t$$

For an ideal gas, pressure is related to molecular motion by:

$$PV = Nk_B T$$

Average kinetic energy of one molecule:

$$K = \frac{3}{2} k_B T$$

Total kinetic energy of an ideal gas:

$$K = \frac{3}{2} Nk_B T$$

Mean squared speed of gas molecules:

$$v^2 = 3k_B T / m$$

Heating causes changes of state (phase changes).

Evaporation occurs because fast-moving molecules escape from the liquid surface.

During evaporation, energetic molecules leave after gaining sufficient kinetic energy through molecular collisions.

Latent Heat of Fusion:

$$Q = mL_F$$

Latent Heat of Vaporization:

$$Q = mL_V$$

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Heat required to change temperature:

$$Q = mc\Delta T$$

During melting and boiling, **temperature remains constant** until the phase change is complete.

Entropy measures the **degree of disorder** in a system.

Entropy change for a small heat transfer:

$$dS = dQ/T$$

Entropy increases when heat is added.

Entropy also increases when the volume of a gas increases.

Entropy change for an ideal gas:

$$\Delta S = C_V \ln(T_2/T_1) + Nk_B \ln(V_2/V_1)$$

Probability of finding a particle with speed v_i :

$$P(v_i) = n(v_i)/N$$

Average speed:

$$v = \sum v_i P(v_i)$$

Statistical mechanics always deals with **average values** rather than individual particles.

20 Difficult Statement, Formula & Numerical MCQs

1. Heat is best defined as:

- A) Potential energy of atoms
- B) Electrical energy
- C) Random kinetic energy of atoms and molecules ✓
- D) Chemical energy

2. The branch of physics that explains heat using molecular motion is called:

- A) Thermodynamics
- B) Electromagnetism

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

C) Statistical Mechanics ✓

D) Optics

3. The equation of state for an ideal gas is:

A) $PV = nRT^2$

B) $P = V/T$

C) $PV = Nk_B T$ ✓

D) $PV = Nk_B/T$

4. At constant pressure, the volume of an ideal gas is:

A) Inversely proportional to temperature

B) Directly proportional to temperature ✓

C) Independent of temperature

D) Equal to pressure

5. Pressure is defined as:

A) Force $\times$ Area

B) Force / Area ✓

C) Area / Force

D) Force² / Area

6. According to Newton's Second Law:

A) $F = ma^2$

B) $F = mv$

C) $F = \Delta p / \Delta t$ ✓

D) $F = p/t^2$

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

7. The average kinetic energy of one gas molecule is:

- A) $k_B T$
- B) $2k_B T$
- C) $\frac{3}{2} k_B T$ ✓
- D) $\frac{5}{2} k_B T$

8. The total kinetic energy of an ideal gas is:

- A) $Nk_B T$
- B) $Nk_B T/2$
- C) $\frac{3}{2} Nk_B T$ ✓
- D) $\frac{5}{2} Nk_B T$

9. The mean squared speed of gas molecules is:

- A) $m/3k_B T$
- B) $2k_B T/m$
- C) $3k_B T/m$ ✓
- D) $k_B T/3m$

10. Evaporation occurs because:

- A) Molecules lose kinetic energy.
- B) Molecules stop moving.
- C) Fast-moving molecules escape from the liquid surface. ✓
- D) Pressure becomes zero.

11. Heat required for melting is calculated by:

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

A) $Q = mc\Delta T$

B) $Q = mL(F)$ ✓

C) $Q = mL(V)$

D) $Q = P\Delta V$

12. Heat required for vaporization is:

A) $Q = mc\Delta T$

B) $Q = mL(F)$

C) $Q = mL(V)$ ✓

D) $Q = PV$

13. Heat required to raise temperature is given by:

A) $Q = mL(F)$

B) $Q = mL(V)$

C) $Q = mc\Delta T$ ✓

D) $Q = PT$

14. During melting or boiling, the temperature:

A) Continuously increases

B) Continuously decreases

C) Remains constant until the phase change is complete ✓

D) Becomes zero

15. Entropy is a measure of:

A) Pressure

B) Density

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

C) Degree of disorder ✓

D) Elasticity

16. The basic entropy relation is:

A) $dS = TdQ$

B) $dS = QT$

C) $dS = dQ/T$ ✓

D) $dS = T/Q$

17. Entropy of a gas increases when:

A) Temperature decreases

B) Volume decreases

C) Temperature or volume increases ✓

D) Pressure becomes zero

18. The probability of finding a particle with speed v_i is:

A) $P(v_i) = N/n(v_i)$

B) $P(v_i) = n(v_i)/N$ ✓

C) $P(v_i) = N \times n(v_i)$

D) $P(v_i) = v_i/N$

19. The average speed of particles is calculated using:

A) $v = \Sigma P(v_i)$

B) $v = \Sigma n(v_i)$

C) $v = \Sigma v_i P(v_i)$ ✓

D) $v = P/N$

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

20. If the temperature of an ideal gas doubles while the mass of each molecule remains constant, the mean squared speed of the molecules becomes:

- A) One-half of its original value
- B) Twice its original value ✓
- C) Four times its original value
- D) Unchanged

Lesson 39: Special Relativity I

Important Exam Points

Special Theory of Relativity was introduced by **Albert Einstein in 1905**.

Special Relativity explains the relationship between **space, time, mass, and energy**.

Einstein's General Theory of Relativity deals with **gravity**, while Special Relativity deals with **uniform motion**.

Time is measured using repeating phenomena such as:

Heart beats, pendulums, quartz crystals, Earth's rotation, and atomic clocks.

Atomic clocks are extremely accurate and can measure time up to one part in a trillion.

Distance is measured by comparing an object with a standard unit of length such as a meter.

Newton believed that:

Time is absolute and the same everywhere.

Space is absolute and independent of the observer.

Einstein showed that:

Time and space are not absolute quantities.

They depend on the motion of the observer.

An **event** is something that occurs at a particular position and time.

Galilean transformations are:

$$x' = x - vt$$

$$y' = y$$

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

$$z' = z$$

$$t' = t$$

Galilean transformations assume that time is the same for all observers.

According to Galilean velocity transformation:

$$u'_x = u_x - v$$

$$u'_y = u_y$$

$$u'_z = u_z$$

Acceleration remains the same in Galilean transformation.

Light is an electromagnetic wave.

The speed of light in vacuum is:

$$c = 3 \times 10^8 \text{ m/s}$$

The old concept of **aether** was rejected because no evidence was found for it.

Einstein proposed two postulates of Special Relativity:

The laws of physics are the same in all inertial frames.

The speed of light in vacuum is constant for all observers regardless of source or observer motion.

Moving clocks run slower compared to stationary clocks.

This phenomenon is called **Time Dilation**.

Time dilation formula:

$$\Delta t = \gamma \Delta t'$$

where:

$$\gamma = 1 / \sqrt{1 - v^2/c^2}$$

The value of γ is always greater than 1.

As velocity approaches the speed of light, time dilation increases.

The lifetime of unstable particles such as **muons** increases due to time dilation.

Objects moving at high speeds become shorter in the direction of motion.

This effect is called **Length Contraction**.

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Length contraction formula:

$$\Delta x = \Delta x' / \gamma$$

Only the dimension parallel to motion contracts.

Dimensions perpendicular to motion remain unchanged.

Lorentz Transformation replaces Galilean Transformation in Special Relativity.

Lorentz Transformation for position:

$$x' = \gamma(x - vt)$$

Lorentz Transformation for time:

$$t' = \gamma(t - vx/c^2)$$

When velocity is very small compared to the speed of light:

$$\gamma \approx 1$$

and Lorentz transformation becomes Galilean transformation.

20 Difficult Statement, Formula & Numerical MCQs

1. The Special Theory of Relativity was proposed by:

- A) Newton
- B) Maxwell
- C) Albert Einstein ✓**
- D) Galileo

2. Special Relativity was introduced in the year:

- A) 1895
- B) 1905 ✓**
- C) 1915
- D) 1925

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

3. Special Relativity mainly deals with:

- A) Electricity and magnetism
 - B) Gravity only
 - C) Space, time, and motion ✓
 - D) Chemical reactions
-

4. Newton believed that time was:

- A) Relative
 - B) Absolute ✓
 - C) Variable
 - D) Dependent on speed
-

5. An event in relativity is defined as:

- A) A moving object
 - B) A wave phenomenon
 - C) Something occurring at a specific position and time ✓
 - D) A force acting on an object
-

6. The Galilean transformation for x-coordinate is:

- A) $x' = x + vt$
 - B) $x' = x - vt$ ✓
 - C) $x' = vt - x$
 - D) $x' = \gamma x$
-

7. Galilean transformation assumes that:

- A) Light speed changes

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

B) Time is the same in all reference frames ✓

C) Space does not exist

D) Mass changes

8. According to Galilean velocity transformation:

A) $u'_x = u_x + v$

B) $u'_x = u_x - v$ ✓

C) $u'_x = vu_x$

D) $u'_x = u_x/v$

9. The speed of light in vacuum is approximately:

A) 3×10^6 m/s

B) 3×10^7 m/s

C) 3×10^8 m/s ✓

D) 3×10^{10} m/s

10. Einstein rejected the concept of:

A) Gravity

B) Energy

C) Aether ✓

D) Motion

11. One postulate of Special Relativity states that:

A) Time is absolute

B) Laws of physics are same in all inertial frames ✓

C) Mass is always constant

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

D) Space has no dimensions

12. The second postulate of Special Relativity states that light speed:

- A) Depends on observer speed
 - B) Changes with source motion
 - C) Is constant for all observers ✓**
 - D) Is zero in vacuum
-

13. The phenomenon in which moving clocks run slower is called:

- A) Length contraction
 - B) Time dilation ✓**
 - C) Relativistic mass
 - D) Doppler effect
-

14. The time dilation formula is:

- A) $\Delta t = \Delta t' / \gamma$
 - B) $\Delta t = \gamma \Delta t'$ ✓**
 - C) $\Delta t = \gamma + \Delta t'$
 - D) $\Delta t = \gamma^2 \Delta t'$
-

15. The Lorentz factor is given by:

- A) $\gamma = 1 - v^2/c^2$
 - B) $\gamma = 1/\sqrt{1 - v^2/c^2}$ ✓**
 - C) $\gamma = v/c$
 - D) $\gamma = c/v$
-

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

16. As velocity approaches the speed of light, the value of γ :

- A) Becomes zero
 - B) Remains one
 - C) Increases greatly ✓
 - D) Becomes negative
-

17. Length contraction occurs:

- A) In all directions equally
 - B) Only along the direction of motion ✓
 - C) Only perpendicular to motion
 - D) Only at rest
-

18. The length contraction formula is:

- A) $\Delta x = \gamma \Delta x'$
 - B) $\Delta x = \Delta x' / \gamma$ ✓
 - C) $\Delta x = \gamma^2 \Delta x'$
 - D) $\Delta x = \Delta x' + \gamma$
-

19. The relativistic transformation replacing Galilean transformation is:

- A) Newton Transformation
 - B) Maxwell Transformation
 - C) Lorentz Transformation ✓
 - D) Einstein Equation
-

20. When velocity is much smaller than the speed of light, Lorentz transformation becomes:

- A) Quantum transformation

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

B) Galilean transformation ✓

C) Energy transformation

D) Photon transformation

Lesson 40: Special Relativity II

Important Exam Points

Lorentz Transformation relates the measurements of **space and time** between two inertial reference frames.

The Lorentz transformations are:

$$x' = \gamma(x - vt)$$

$$t' = \gamma(t - vx/c^2)$$

where:

$$\gamma = 1 / \sqrt{1 - v^2/c^2}$$

For two events, the space and time intervals transform as:

$$\Delta x' = \gamma(\Delta x - v\Delta t)$$

$$\Delta t' = \gamma(\Delta t - v\Delta x/c^2)$$

Einstein's velocity addition formula is:

$$u' = (u - v) / (1 - uv/c^2)$$

This formula replaces the classical Galilean velocity transformation at high speeds.

At low velocities compared with the speed of light:

$$u \ll c$$

Einstein velocity addition becomes:

$$u' = u - v$$

which is the classical Galilean result.

According to Special Relativity, the speed of light remains constant.

If a car moves with speed v and turns on its headlight, the observer still measures:

Speed of light = c

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

The relativistic velocity formula proves that the motion of the source does not affect the speed of light.

The space-time interval is defined as:

$$I = (c\Delta t)^2 - (\Delta x)^2$$

The interval remains unchanged for all inertial observers.

This property is called **invariance of space-time interval**.

If:

$$I = 0$$

the interval is called **light-like interval**.

If:

$$I > 0$$

the interval is called **time-like interval**.

If:

$$I < 0$$

the interval is called **space-like interval**.

For a time-like interval, all observers agree about the order of events.

For a space-like interval, different observers may disagree about the order of events.

The path of an object through space-time is called its **world line**.

A stationary object has a vertical world line.

A moving object has a tilted world line.

A photon always moves along the light cone because it travels at speed c .

The light cone separates:

Future events

Past events

Events that cannot influence each other.

The relativistic Doppler effect explains the change in frequency of light due to relative motion.

When the source approaches an observer:

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Frequency increases.

When the source moves away:

Frequency decreases.

Relativistic Doppler formula:

$$v = \sqrt{\frac{(c+v)}{(c-v)}} v_0$$

The Doppler effect is used to measure the speed of stars and other moving objects.

Newtonian momentum:

$$p = mu$$

is not valid at relativistic speeds.

The relativistic momentum is:

$$p = \gamma mu$$

At low speeds:

$$\gamma \approx 1$$

and relativistic momentum becomes classical momentum.

Relativistic momentum follows the conservation law:

$$m_A \gamma_A u_A + m_B \gamma_B u_B = m_C \gamma_C u_C + m_D \gamma_D u_D$$

Classical kinetic energy:

$$K = \frac{1}{2} mu^2$$

is not valid for objects moving near the speed of light.

The relativistic energy equation is:

$$E = \gamma mc^2$$

The rest energy of an object is:

$$E_0 = mc^2$$

The relationship between kinetic energy and total energy is:

$$K = E - E_0$$

At low velocities, relativistic energy becomes:

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$$E \approx mc^2 + \frac{1}{2}mu^2$$

The energy-momentum relationship is:

$$E^2 = p^2c^2 + m^2c^4$$

For massless particles like photons:

$$m = 0$$

Therefore:

$$E = pc$$

Einstein's famous equation:

$$E = mc^2$$

shows the equivalence of mass and energy.

A small amount of mass can produce a very large amount of energy because the speed of light is extremely large.

20 Difficult Statement, Formula & Numerical MCQs

1. Lorentz transformation is used to relate:

- A) Force and acceleration
 - B) Mass and energy only
 - C) Space and time between inertial frames ✓
 - D) Temperature and heat
-

2. The Lorentz transformation for position is:

- A) $x' = x + vt$
 - B) $x' = \gamma(x - vt)$ ✓
 - C) $x' = \gamma(x + vt)$
 - D) $x' = x/v$
-

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

3. The Lorentz factor is represented by:

- A) β
 - B) γ ✓
 - C) λ
 - D) μ
-

4. Einstein velocity addition formula is:

- A) $u' = u + v$
 - B) $u' = (u - v)/(1 - uv/c^2)$ ✓
 - C) $u' = uv$
 - D) $u' = u/v$
-

5. When velocity is much smaller than the speed of light, Einstein velocity addition becomes:

- A) $u' = c$
 - B) $u' = u - v$ ✓
 - C) $u' = uv$
 - D) $u' = v/u$
-

6. According to Special Relativity, the speed of light is:

- A) Different for each observer
 - B) Dependent on source motion
 - C) Constant for all observers ✓
 - D) Zero in vacuum
-

7. The invariant quantity in relativity is:

- A) Velocity

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

B) Energy

C) Space-time interval ✓

D) Momentum only

8. The space-time interval is given by:

A) $I = \Delta x + c\Delta t$

B) $I = (c\Delta t)^2 - (\Delta x)^2$ ✓

C) $I = c\Delta x/\Delta t$

D) $I = \Delta x^2 + \Delta t^2$

9. When $I = 0$, the interval is called:

A) Space-like

B) Time-like

C) Light-like ✓

D) Absolute interval

10. For a time-like interval:

A) $I < 0$

B) $I > 0$ ✓

C) $I = 0$

D) $I = 1$

11. The path of an object through space-time is called:

A) Orbit

B) Trajectory only

C) World line ✓

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

D) Wave function

12. A photon moves along the:

A) Time axis

B) Space axis

C) Light cone ✓

D) World surface

13. The relativistic Doppler effect is related to change in:

A) Mass

B) Momentum

C) Frequency of light ✓

D) Temperature

14. Relativistic momentum is given by:

A) $p = mu^2$

B) $p = \gamma mu$ ✓

C) $p = mc^2$

D) $p = m/u$

15. At low speeds relativistic momentum becomes:

A) Zero momentum

B) Classical momentum $p = mu$ ✓

C) Infinite momentum

D) Negative momentum

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

16. The relativistic energy formula is:

- A) $E = \frac{1}{2}mu^2$
 - B) $E = \gamma mc^2$ ✓
 - C) $E = mv$
 - D) $E = pc^2$
-

17. The rest energy of a particle is:

- A) $E = mv^2$
 - B) $E_0 = mc^2$ ✓
 - C) $E = \gamma mu$
 - D) $E = pc$
-

18. The energy-momentum relationship is:

- A) $E = mc$
 - B) $E^2 = p^2c^2 + m^2c^4$ ✓
 - C) $E = p/m$
 - D) $E = m/c^2$
-

19. For a photon, because mass is zero:

- A) $E = mc^2$
 - B) $E = 0$
 - C) $E = pc$ ✓
 - D) $E = p/c^2$
-

20. Einstein's equation $E = mc^2$ shows:

- A) Energy disappears

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

B) Speed affects mass only

C) Mass and energy are equivalent ✓

D) Light has no energy

Lesson 41: Waves and Particles

Important Exam Points

Wave-particle duality states that matter and light can behave both as **waves and particles** depending on the situation.

Light shows particle nature through experiments like:

Photoelectric Effect and Compton Effect.

According to classical physics, increasing light intensity should increase the kinetic energy of emitted electrons, but experiments showed that:

The maximum kinetic energy of electrons depends on **frequency of light**, not intensity.

Electrons are emitted only when light frequency is above a certain minimum value called **threshold frequency**.

Einstein explained the photoelectric effect by proposing that light consists of small packets of energy called **photons**.

Energy of a photon is:

$$E = h\nu$$

where:

$$h = \text{Planck's constant} = 6.63 \times 10^{-34} \text{ Js}$$

$$\nu = \text{frequency of light}$$

The work function (W) is the minimum energy required to remove an electron from a material.

The maximum kinetic energy of emitted electrons is:

$$K_{\text{max}} = h\nu - W$$

Arthur Compton experimentally proved the particle nature of light through the **Compton Effect**.

Compton Effect explains the change in wavelength of scattered photons after collision with electrons.

Compton wavelength shift formula:

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

$$\Delta\lambda = \lambda' - \lambda = (h/mec)(1 - \cos\theta)$$

The Compton wavelength is:

$$\lambda_c = h/mec = 2.4 \times 10^{-12} \text{ m}$$

The frequency of scattered photons decreases because photons transfer energy and momentum to electrons.

Important photon properties:

Photon energy:

$$E = h\nu$$

Photon momentum:

$$p = h/\lambda$$

Energy-momentum relation for photons:

$$E = pc$$

Relation between wavelength, frequency, and speed:

$$\lambda\nu = c$$

Light is always detected in the form of complete packets called photons.

Louis de Broglie proposed that matter particles also have wave properties.

The wavelength associated with a particle is called **de Broglie wavelength**.

De Broglie wavelength formula:

$$\lambda = h/p$$

where:

p = momentum of particle

Large objects like cricket balls have extremely small wavelengths, while electrons have wavelengths comparable to atomic dimensions.

The Davisson-Germer experiment confirmed the wave nature of electrons.

Electron diffraction occurs because electron waves produce:

Interference and diffraction patterns

The double-slit experiment shows that:

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Light, electrons, and atoms all behave as waves.

Matter waves depend on momentum, not the size of particles.

A single particle can interfere with itself.

If we try to determine which slit a particle passes through, the interference pattern disappears.

This proves that wave and particle behavior cannot be observed simultaneously.

Heisenberg introduced the **Uncertainty Principle**.

It states that the exact position and momentum of a particle cannot be known at the same time.

The uncertainty relation is:

$$\Delta x \Delta p \geq \hbar/2$$

where:

Δx = uncertainty in position

Δp = uncertainty in momentum

If position uncertainty decreases, momentum uncertainty increases.

Quantum uncertainty is a fundamental property of nature, not only due to measurement disturbance.

Energy-time uncertainty principle:

$$\Delta E \Delta t \geq \hbar/2$$

This means a particle can have uncertainty in energy for a very short time.

Because of energy-time uncertainty, atomic energy levels are not perfectly sharp.

Atomic transitions produce a small spread in emitted frequencies.

20 Difficult Statement, Formula & Numerical MCQs

1. Wave-particle duality means that matter and light can behave as:

- A) Only particles
- B) Only waves
- C) Both waves and particles ✓
- D) Neither waves nor particles

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

2. The particle nature of light was confirmed by:

- A) Newton's law
- B) Photoelectric effect ✓**
- C) Reflection
- D) Refraction

3. In the photoelectric effect, electron kinetic energy depends on:

- A) Light intensity only
- B) Frequency of light ✓**
- C) Distance from source
- D) Metal size

4. Einstein explained the photoelectric effect by assuming light consists of:

- A) Waves only
- B) Electric fields
- C) Photons ✓**
- D) Sound particles

5. The energy of a photon is given by:

- A) $E = mc^2$
- B) $E = h\nu$ ✓**
- C) $E = p^2/2m$
- D) $E = mv$

6. The value of Planck's constant is approximately:

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

A) $6.63 \times 10^{-6} \text{ Js}$

B) $6.63 \times 10^{-34} \text{ Js}$ ✓

C) $3 \times 10^8 \text{ Js}$

D) 9.8 Js

7. The minimum energy needed to remove an electron from a metal is called:

A) Photon energy

B) Work function ✓

C) Momentum energy

D) Rest energy

8. The maximum kinetic energy of photoelectrons is:

A) $K_{\max} = h\nu + W$

B) $K_{\max} = h\nu - W$ ✓

C) $K_{\max} = mc^2$

D) $K_{\max} = p\lambda$

9. The Compton effect proves the:

A) Wave nature of electrons

B) Particle nature of light ✓

C) Gravitational nature of light

D) Nuclear nature of light

10. The Compton wavelength shift depends on:

A) Light intensity

B) Frequency only

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

C) Scattering angle ✓

D) Temperature

11. The energy-momentum relation for photons is:

A) $E = mc^2$

B) $E = pc$ ✓

C) $E = mv^2$

D) $E = p^2/2m$

12. Photon momentum is given by:

A) $p = mc$

B) $p = h/\lambda$ ✓

C) $p = mv^2$

D) $p = h\lambda$

13. De Broglie proposed that matter particles have:

A) Only mass

B) Wave properties ✓

C) No energy

D) No momentum

14. The de Broglie wavelength formula is:

A) $\lambda = p/h$

B) $\lambda = h/p$ ✓

C) $\lambda = hc/E$

D) $\lambda = mv/h$

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

15. The wave nature of electrons was confirmed by:

- A) Photoelectric experiment
- B) Davisson-Germer experiment ✓**
- C) Rutherford experiment
- D) Millikan experiment

16. The double-slit experiment demonstrates:

- A) Only particle behavior
- B) Interference of matter waves ✓**
- C) Energy conservation only
- D) Nuclear reactions

17. Heisenberg uncertainty principle relates:

- A) Energy and mass
- B) Position and momentum ✓**
- C) Frequency and speed
- D) Force and acceleration

18. The uncertainty relation is:

- A) $\Delta x \Delta p = 0$
- B) $\Delta x \Delta p \geq h/2$ ✓**
- C) $\Delta x \Delta p = mc^2$
- D) $\Delta x \Delta p = h\nu$

19. If position uncertainty decreases, momentum uncertainty:

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

- A) Decreases
 - B) Increases ✓**
 - C) Becomes zero
 - D) Remains constant
-

20. Energy-time uncertainty principle is:

- A) $\Delta E \Delta t = mc^2$
- B) $\Delta E \Delta t \geq \hbar/2$ ✓**
- C) $\Delta E \Delta t = h\lambda$
- D) $\Delta E \Delta t = pv$

Lesson 42: Quantum Mechanics

Important Exam Points

The word **quantum** means a packet or bundle of energy.

Quantum values are **discrete**, meaning they exist in separate steps rather than continuous values.

Quantum Mechanics is the branch of physics that explains the behavior of the **microscopic world**.

Important microscopic scales:

Atom size:

10^{-10} m

Nucleus size:

10^{-14} m

Nucleon size:

10^{-15} m

Quantum Mechanics is necessary because classical Newtonian mechanics fails at atomic levels.

Classical mechanics works well for large objects, but it cannot explain microscopic phenomena.

The main ideas of Quantum Mechanics are:

Atoms and molecules can exist only in specific energy states called:

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Quantum states or allowed energy levels

Each quantum state is described by quantum numbers.

Atoms absorb or emit energy when they move between energy states.

The energy absorbed or released equals the difference between two energy levels.

Quantum Mechanics always deals with **probabilities** rather than exact predictions.

The Quantum Revolution occurred because classical physics failed to explain:

Blackbody radiation

Photoelectric effect

Stability of atoms and atomic spectra

Blackbody Radiation

Classical physics predicted that radiation energy should increase continuously with frequency.

According to classical theory:

Energy radiation would become infinite at high frequencies.

This problem was called the:

Ultraviolet Catastrophe

Max Planck solved this problem by introducing energy quanta.

Planck proposed that energy is emitted or absorbed in packets:

$$E = h\nu$$

where:

h = Planck constant

ν = frequency

Planck assumed electromagnetic energy consists of finite energy elements.

Photoelectric Effect

Einstein explained the photoelectric effect by proposing that light consists of particles called:

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Photons

A photon has:

Energy

Momentum

A photon removes an electron from a metal only when it has enough energy.

The photon energy is:

$$E = h\nu$$

The minimum energy required to remove an electron is called:

Work Function (W)

Maximum kinetic energy of emitted electrons:

$$K_{\max} = h\nu - W$$

Atomic Stability and Bohr Model

Classical physics could not explain why atoms are stable.

According to classical electromagnetic theory:

Accelerating electrons should continuously lose energy and collapse into the nucleus.

Niels Bohr proposed that electrons can exist only in specific energy levels.

Electron angular momentum is quantized:

$$L = n\hbar$$

where:

$$n = 1, 2, 3, \dots$$

Bohr's model explained why atoms have:

Specific energies

Discrete spectral lines

Uncertainty Principle and Stability of Atoms

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Heisenberg uncertainty principle states that position and momentum cannot both be known exactly at the same time.

Formula:

$$\Delta x \Delta p \geq \hbar/2$$

If an electron is confined to a very small region:

Position uncertainty decreases.

Momentum uncertainty increases.

Higher momentum causes higher kinetic energy.

This prevents electrons from collapsing into the nucleus.

Quantum Harmonic Oscillator

In classical physics, an oscillator can have any energy.

In quantum mechanics, oscillator energy is quantized.

Allowed energy levels are:

$$E = (n + 1/2)\hbar\omega$$

where:

$$n = 0, 1, 2, 3, \dots$$

The lowest possible energy is:

$$E = 1/2 \hbar\omega$$

This is called:

Zero-point energy

Zero-point energy exists because complete rest is impossible according to uncertainty principle.

Quantum Tunneling

Quantum mechanics predicts that particles can pass through energy barriers even when they do not have enough classical energy.

This phenomenon is called:

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Tunneling

According to energy-time uncertainty:

$$\Delta E \Delta t \geq \hbar/2$$

A particle can temporarily borrow energy for a short time and cross a barrier.

Tunneling explains:

Nuclear fusion inside the Sun.

Protons overcome electrostatic repulsion and combine due to tunneling.

Probability in Quantum Mechanics

Quantum mechanics describes events using probability.

Probability measures the likelihood of an event occurring.

Probability range:

0 = impossible event

1 = certain event

Mathematically:

$$P = n/N$$

where:

n = number of successful events

N = total experiments

Quantum States and Amplitudes

A quantum state can be represented as:

$$\Psi = c_1 \uparrow + c_2 \downarrow$$

where:

c_1 and c_2 are quantum amplitudes.

Probability of finding each state:

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

$$P(\uparrow) = |c_1|^2$$

$$P(\downarrow) = |c_2|^2$$

The total probability must be:

$$P(\uparrow) + P(\downarrow) = 1$$

Quantum mechanics adds amplitudes first and then calculates probability.

This produces:

Interference effects

Stern-Gerlach Experiment

The Stern-Gerlach experiment showed that electrons have discrete spin states.

Electrons can exist in:

Spin up state $\uparrow$

Spin down state $\downarrow$

The experiment proved that microscopic particles have quantized properties.

Wave Function

In 1926, Erwin Schrödinger introduced the concept of the:

Wave Function $\Psi(x,t)$

Wave function describes the state of a particle.

The probability of finding a particle is:

$$|\Psi(x,t)|^2 dx$$

The total probability must be equal to:

$$\int |\Psi(x,t)|^2 dx = 1$$

The wave function is obtained by solving:

Schrödinger Equation

The Schrödinger equation provides information about:

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Energy levels

Particle location probability

Momentum

Atomic behavior

20 Difficult Statement, Formula & Numerical MCQs

1. The word quantum means:

- A) Continuous energy
 - B) Packet or bundle of energy ✓**
 - C) Force
 - D) Motion
-

2. Quantum Mechanics mainly describes:

- A) Large objects only
 - B) Microscopic world ✓**
 - C) Planetary motion
 - D) Classical waves
-

3. The approximate size of an atom is:

- A) 10^{-5} m
 - B) 10^{-10} m ✓**
 - C) 10^{-14} m
 - D) 10^{-20} m
-

4. Classical mechanics fails at atomic scale because of:

- A) Gravity

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

B) Uncertainty principle ✓

C) Temperature

D) Friction

5. Allowed energy states of atoms are called:

A) Force levels

B) Quantum states ✓

C) Thermal states

D) Motion states

6. Quantum Mechanics deals mainly with:

A) Exact predictions

B) Probabilities ✓

C) Fixed paths

D) Continuous energy

7. The three major problems that caused quantum revolution included:

A) Gravity and motion

B) Blackbody radiation, photoelectric effect, atomic stability ✓

C) Heat and sound

D) Magnetism only

8. Planck suggested that energy is emitted in the form of:

A) Waves only

B) Quanta ✓

C) Forces

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

D) Fields

9. The energy of a quantum is:

A) $E = mc^2$

B) $E = h\nu$ ✓

C) $E = mv^2$

D) $E = p^2/2m$

10. Einstein explained the photoelectric effect using:

A) Electrons

B) Photons ✓

C) Neutrons

D) Waves only

11. Bohr proposed that electrons exist in:

A) Random energies

B) Quantized energy levels ✓

C) Continuous energy states

D) Zero energy states

12. Heisenberg uncertainty principle relates:

A) Energy and time only

B) Position and momentum ✓

C) Mass and energy

D) Frequency and wavelength

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

13. The uncertainty principle formula is:

- A) $\Delta x \Delta p = mc^2$
 - B) $\Delta x \Delta p \geq \hbar/2$ ✓
 - C) $\Delta x \Delta p = h\nu$
 - D) $\Delta x \Delta p = 1$
-

14. Zero-point energy is associated with:

- A) Classical motion
 - B) Quantum harmonic oscillator ✓
 - C) Gravity
 - D) Nuclear decay
-

15. The lowest energy of quantum oscillator is:

- A) Zero
 - B) $1/2\hbar\omega$ ✓
 - C) $\hbar\omega$
 - D) $2\hbar\omega$
-

16. Quantum tunneling occurs because of:

- A) Newton's laws
 - B) Uncertainty principle ✓
 - C) Gravity
 - D) Friction
-

17. The Sun's fusion process is possible partly due to:

- A) Reflection

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

B) Quantum tunneling ✓

C) Cooling

D) Radiation loss

18. Probability values in quantum mechanics range from:

A) -1 to 1

B) 0 to 1 ✓

C) 1 to 10

D) 10 to 100

19. The probability of a quantum state is determined by:

A) Wave function only

B) Square of amplitude ✓

C) Energy only

D) Momentum only

20. Schrödinger equation determines:

A) Gravitational force

B) Wave function of a particle ✓

C) Speed of light

D) Classical motion only

Lesson 43: Introduction to Atomic Physics

Important Exam Points

Ancient Greek philosopher **Democritus** proposed that the world is made of tiny indivisible particles called **atoms**.

The word **atom** means indivisible particle.

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Lavoisier introduced the **Law of Conservation of Matter**, which states that mass remains the same before and after a chemical reaction.

Dalton (1803) presented the atomic theory:

Atoms are the basic building blocks of elements.

All atoms of the same element have the same mass.

Atoms of different elements have different properties.

Different atoms combine in simple ratios to form compounds.

Avogadro's hypothesis states that equal volumes of all gases at the same temperature and pressure contain equal numbers of molecules.

The size of an atom is approximately:

10^{-10} m

Atoms are mostly empty space.

J.J. Thomson (1895) proposed the **Plum Pudding Model** of the atom.

According to Thomson:

The atom contains positively charged material.

Electrons are scattered inside this positive charge.

Thomson's model was later rejected.

Rutherford's Experiment (1911) used alpha particles and gold foil to study atomic structure.

Rutherford discovered:

A small, dense, positively charged nucleus exists inside the atom.

Most of the atom is empty space.

Electrons move around the nucleus.

Rutherford's model was similar to the solar system model.

According to classical physics, electrons moving around the nucleus should continuously radiate energy.

Because of energy loss, electrons should fall into the nucleus.

This problem showed that classical physics could not explain atomic stability.

Atomic spectra also created problems for classical physics.

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Atoms do not emit continuous light spectra.

Atoms emit light at specific discrete frequencies called **spectral lines**.

Bohr introduced his atomic model to explain atomic spectra.

Bohr proposed that electrons exist only in certain allowed orbits.

These allowed orbits have fixed energies.

Electrons do not radiate energy while staying in an allowed orbit.

Electrons emit or absorb photons when they move between energy levels.

The energy of photon is:

$$h\nu = E_f - E_i$$

where:

h = Planck constant

ν = frequency of photon

Bohr used the wave nature of electrons.

According to De Broglie:

$$\lambda = h/p$$

Electrons behave as waves.

Only those orbits are allowed where electron waves form standing waves.

The angular momentum of electron is quantized:

$$L = n\hbar$$

where:

$$n = 1, 2, 3, \dots$$

The radius of Bohr orbit is:

$$r_n = a_0 n^2$$

where:

a_0 = Bohr radius

The Bohr radius is:

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

$$a_0 = 0.53 \times 10^{-10} \text{ m}$$

For hydrogen atom, energy levels are:

$$E_n = -13.6/n^2 \text{ eV}$$

Negative energy shows that the electron is bound to the nucleus.

The ground state is:

$$n = 1$$

Higher values of n represent excited states.

Bohr model successfully explained the hydrogen spectrum.

However, it failed for multi-electron atoms.

Quantum Mechanics provides the correct description of atomic structure.

According to Quantum Mechanics:

Electrons are described by wave functions.

The square of wave function gives the probability of finding an electron.

$|\psi|^2$ represents electron probability density.

The Schrödinger equation determines possible electron states.

The principal quantum number:

$$n = 1, 2, 3, \dots$$

determines energy levels.

Orbital angular momentum is determined by:

$$l = 0, 1, 2, \dots, n-1$$

The magnetic quantum number:

$$m_l$$

determines the direction of angular momentum.

Electron spin is described by:

$$m_s = +1/2 \text{ or } -1/2$$

Electrons behave like tiny magnets due to their spin.

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Atomic orbitals are named:

s, p, d, f

s-orbitals are spherical in shape.

20 Difficult Statement, Formula & Numerical MCQs

1. The concept of atoms was first proposed by:

A) Newton

B) Einstein

C) Democritus ✓

D) Rutherford

2. The Law of Conservation of Matter was introduced by:

A) Dalton

B) Lavoisier ✓

C) Bohr

D) Thomson

3. Dalton proposed atomic theory in:

A) 1700

B) 1803 ✓

C) 1905

D) 1911

4. According to Avogadro's hypothesis, equal volumes of gases contain equal numbers of molecules when they have the same:

A) Mass only

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

B) Temperature and pressure ✓

C) Density only

D) Volume only

5. The approximate size of an atom is:

A) 10^{-5} m

B) 10^{-8} m

C) 10^{-10} m ✓

D) 10^{-20} m

6. The Plum Pudding Model was proposed by:

A) Rutherford

B) J.J. Thomson ✓

C) Bohr

D) Dalton

7. Rutherford discovered the:

A) Electron

B) Photon

C) Atomic nucleus ✓

D) Neutron

8. Rutherford used which particles in his experiment?

A) Beta particles

B) Alpha particles ✓

C) Electrons

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

D) Photons

9. According to classical physics, accelerating charges:

A) Gain mass

B) Radiate energy ✓

C) Stop moving

D) Become neutral

10. Atomic spectra consist of:

A) Continuous colors only

B) Discrete frequencies ✓

C) Random frequencies

D) No frequencies

11. Bohr proposed that electrons move in:

A) Random paths

B) Allowed energy levels ✓

C) Straight lines

D) No orbits

12. The energy of a photon is given by:

A) $E = mc^2$

B) $E = h\nu$ ✓

C) $E = mv^2$

D) $E = p^2/2m$

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

13. De Broglie wavelength formula is:

A) $\lambda = pc$

B) $\lambda = h/p$ ✓

C) $\lambda = p/h$

D) $\lambda = hE$

14. The angular momentum in Bohr model is:

A) $L = mv$

B) $L = n\hbar$ ✓

C) $L = mc^2$

D) $L = hv$

15. The Bohr radius is approximately:

A) $0.53 \times 10^{-5} \text{ m}$

B) $0.53 \times 10^{-10} \text{ m}$ ✓

C) $5.3 \times 10^{-2} \text{ m}$

D) 10^{-20} m

16. The energy of hydrogen atom is:

A) $E_n = -13.6n^2$

B) $E_n = -13.6/n^2 \text{ eV}$ ✓

C) $E_n = 13.6n$

D) $E_n = mc^2$

17. The ground state of hydrogen atom corresponds to:

A) $n = 0$

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

B) $n = 1$ ✓

C) $n = 2$

D) $n = 10$

18. The square of wave function represents:

A) Energy

B) Momentum

C) Probability density ✓

D) Charge

19. The principal quantum number determines:

A) Spin direction

B) Energy level ✓

C) Magnetic field

D) Electron charge

20. Electron spin quantum number values are:

A) 0 and 1

B) -1 and $+1$

C) $+1/2$ and $-1/2$ ✓

D) $1/2$ and $2/2$

Lesson 44: Introduction to Nuclear Physics

Important Exam Points

The atom consists of a small, heavy nucleus surrounded by electrons.

The nucleus contains:

Protons

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Neutrons

Both protons and neutrons are called **nucleons**.

The mass of proton and neutron is almost the same and they are approximately **2000 times heavier than an electron**.

The charge of proton is:

$$+1.6 \times 10^{-19} \text{ C}$$

The charge of electron is:

$$-1.6 \times 10^{-19} \text{ C}$$

The neutron has:

No charge

Particle masses are expressed using rest energy:

$$E = mc^2$$

Mass of proton:

$$M_p = 938 \text{ MeV}/c^2$$

Mass of neutron:

$$M_n = 940 \text{ MeV}/c^2$$

Mass of electron:

$$M_e = 0.5 \text{ MeV}/c^2$$

A hydrogen nucleus contains:

One proton only

A deuteron contains:

One proton + one neutron

A triton contains:

One proton + two neutrons

Hydrogen, deuterium, and tritium are called:

Isotopes

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Isotopes have:

Same number of protons

Different numbers of neutrons

Same chemical properties because they have the same number of electrons.

The nuclear notation is:

A_ZX

Where:

Z = atomic number (number of protons)

A = mass number (total nucleons)

N = number of neutrons

The relation between mass number and nucleons is:

$$A = Z + N$$

The approximate size of a nucleus is:

$$10^{-15} \text{ m}$$

The nuclear radius is given by:

$$r = r_0 A^{1/3}$$

where:

$$r_0 = 1.2 \text{ fm}$$

A = mass number

The size of the nucleus increases slowly with increasing mass number.

The nucleus is about:

10 million times smaller than the atom

Electron scattering experiments are used to study nuclear structure.

Electrons interact with protons because both have electric charge.

Neutrons cannot be detected directly through electromagnetic interaction.

To observe the nucleus, electrons must have very small wavelength.

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

The De Broglie relation is:

$$\lambda = h/p$$

A wavelength of approximately:

1 fm

is required to observe the nucleus.

Nuclear density is approximately constant throughout the nucleus.

The density of nucleons is:

0.14 nucleons/fm³

The nucleus remains stable because of the:

Strong Nuclear Force

Properties of strong nuclear force:

It is attractive between nucleons.

It has a very short range.

Its range is approximately:

1–2 fm

It acts between:

Proton-Proton

Neutron-Neutron

Proton-Neutron

At very small distances, the nuclear force becomes repulsive.

In 1935, **Hideki Yukawa** proposed that nuclear force is carried by particles.

He predicted the existence of:

Pion (π -meson)

Yukawa used:

Energy-Time Uncertainty Principle

to predict pion mass.

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

The predicted pion mass was:

124 MeV/c²

Experimental pion mass:

138 MeV/c²

Pions are carriers of the strong nuclear force.

Types of pions are:

π^+

π^0

π^-

Most nuclei are unstable and undergo radioactive decay.

The radioactive decay law is:

$$N = N_0 e^{-\lambda t}$$

where:

N = remaining nuclei

N₀ = initial nuclei

λ = decay constant

Half-life is the time required for half of the radioactive nuclei to decay.

The half-life formula is:

$$T_{1/2} = 0.693/\lambda$$

A larger decay constant means:

More unstable nucleus

Radioactive dating uses radioactive isotopes to determine the age of objects.

Carbon dating uses:

Carbon-14 (¹⁴C)

The half-life of carbon-14 is:

5730 years

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Carbon dating measures the decrease of carbon-14 after an organism dies.

Einstein's equation:

$$E = mc^2$$

is the basis of nuclear energy.

In nuclear fission:

A heavy nucleus breaks into smaller nuclei.

The mass difference is converted into energy.

The energy released is:

$$Q = (M - M_1 - M_2)c^2$$

No completely stable nuclei exist above:

$$Z = 82$$

Binding energy is the energy required to remove a nucleon from a nucleus.

A nucleus with greater binding energy per nucleon is more stable.

The most stable nucleus is:

Iron-56 (^{56}Fe)

Its binding energy is approximately:

8.6 MeV per nucleon

Unstable nuclei can emit:

Alpha (α) radiation

Beta (β) radiation

Gamma (γ) radiation

Alpha particles are:

Helium nuclei

Contain:

2 protons + 2 neutrons

During alpha decay:

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

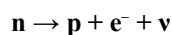
Mass number decreases by 4.

Atomic number decreases by 2.

Beta decay occurs when:

A neutron changes into a proton.

The reaction is:



Beta decay involves:

Weak Nuclear Force

Types of beta decay:

β^- decay $\rightarrow$ electron emission

β^+ decay $\rightarrow$ positron emission

Gamma rays are:

High-energy photons

They are emitted when a nucleus changes from a higher energy state to a lower energy state.

Nuclear energy levels are much larger than atomic energy levels, so gamma photons have very high energy.

20 Difficult Statement, Formula & Numerical MCQs

1. The particles present inside the nucleus are called:

A) Electrons

B) Nucleons ✓

C) Photons

D) Mesons

2. A nucleon includes:

A) Electron and proton

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B) Proton and neutron ✓

C) Photon and electron

D) Neutron and photon

3. The charge of a proton is:

A) $-1.6 \times 10^{-19} \text{ C}$

B) $+1.6 \times 10^{-19} \text{ C}$ ✓

C) Zero

D) $+9.1 \times 10^{-31} \text{ C}$

4. The mass of proton in energy units is approximately:

A) $0.5 \text{ MeV}/c^2$

B) $938 \text{ MeV}/c^2$ ✓

C) $940 \text{ eV}/c^2$

D) $13.6 \text{ MeV}/c^2$

5. A deuteron contains:

A) Two protons

B) One proton and one neutron ✓

C) Two electrons

D) One neutron only

6. Isotopes have the same number of:

A) Neutrons

B) Protons ✓

C) Mass number

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D) Nuclear radius

7. The nuclear notation A_ZX represents:

- A) Electron number only
 - B) Atomic number and mass number ✓**
 - C) Energy levels
 - D) Nuclear force
-

8. The approximate size of a nucleus is:

- A) 10^{-3} m
 - B) 10^{-10} m
 - C) 10^{-15} m ✓**
 - D) 10^{-20} m
-

9. Nuclear radius depends on:

- A) Electron number
 - B) Mass number A ✓**
 - C) Temperature
 - D) Pressure
-

10. The nuclear radius formula is:

- A) $r = r_0 A$
 - B) $r = r_0 A^{1/3}$ ✓**
 - C) $r = mc^2$
 - D) $r = h/p$
-

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11. The strong nuclear force acts over a distance of about:

- A) 1 meter
 - B) 1 kilometer
 - C) 1–2 fm ✓
 - D) Infinite distance
-

12. Yukawa predicted the existence of:

- A) Electron
 - B) Photon
 - C) Pion ✓
 - D) Proton
-

13. The radioactive decay law is:

- A) $E = mc^2$
 - B) $N = N_0 e^{-\lambda t}$ ✓
 - C) $\lambda = h/p$
 - D) $F = ma$
-

14. The half-life formula is:

- A) $T = \lambda/0.693$
 - B) $T_{1/2} = 0.693/\lambda$ ✓
 - C) $T = mc^2$
 - D) $T = h\nu$
-

15. Carbon dating uses the isotope:

- A) ^{12}C

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B) ^{14}C ✓

C) ^{16}O

D) ^{238}U

16. The half-life of carbon-14 is:

A) 100 years

B) 1000 years

C) 5730 years ✓

D) 1 million years

17. Nuclear fission is the process in which:

A) Small nuclei combine

B) Heavy nucleus breaks into smaller nuclei ✓

C) Electrons are emitted

D) Photons disappear

18. The most stable nucleus is:

A) Uranium-235

B) Hydrogen

C) Iron-56 ✓

D) Carbon-14

19. Alpha particles are:

A) Electrons

B) Photons

C) Helium nuclei ✓

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

D) Neutrons

20. Gamma radiation consists of:

A) Electrons

B) Protons

C) High-energy photons ✓

D) Neutrons

Lesson 45: Physics of the Sun

Important Exam Points

The Sun is the main source of energy for life on Earth.

Physics explains the working, energy production, and future evolution of the Sun.

Basic Facts About the Sun

Mass of Sun:

2×10^{30} kg

It is approximately:

333,000 times the mass of Earth

Diameter of Sun:

1,392,000 km

It is approximately:

109 times the diameter of Earth

Age of Sun:

4.6 billion years

Rotation period:

25 days at equator

36 days at poles

Temperature:

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

Core temperature:

15 million K

Surface temperature:

5770 K

Density:

Core density is about:

8 times the density of gold

Average density:

1.5 times the density of water

Composition of Sun:

Hydrogen:

72%

Helium:

25%

Other metals:

3%

Solar Energy and Luminosity

The Sun produces an enormous amount of energy.

Solar luminosity:

3.83×10^{27} joules/second

Power output:

3.83×10^{24} kilowatts

The Sun produces energy equal to billions of Earth power plants.

Chemical fuels cannot explain the Sun's energy because they would last only millions of years.

The real source of solar energy was discovered after Einstein's equation:

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$$E = mc^2$$

The Sun produces energy through:

Nuclear Fusion

Fusion is the process in which atomic nuclei combine to form a heavier nucleus.

Hydrogen Fusion in the Sun

The Sun converts hydrogen nuclei into helium.

Four protons combine to form one helium nucleus.

The mass of helium nucleus is less than the combined mass of four protons.

The missing mass is converted into energy according to:

$$E = mc^2$$

Every second, the Sun converts approximately:

4 million tons of hydrogen into energy

Fusion requires:

High temperature

High density

High pressure

The core temperature of the Sun is high enough for hydrogen fusion.

Hydrogen fusion occurs first because helium fusion requires much higher temperature.

Helium fusion requires approximately:

100 million K

Structure of the Sun

Fusion occurs only in the core of the Sun.

The Sun has different regions:

Core

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

The region where nuclear fusion occurs.

Radiation Zone

Energy moves outward through photons.

Convection Zone

Hot gases rise and cooler gases sink, transferring heat.

Large columns of gas move through convection currents.

Equilibrium of the Sun

The Sun remains stable because two opposite forces balance each other.

1. Gravitational Force

Gravity pulls material inward.

2. Gas Pressure

Hot gases produce pressure that pushes outward.

The balance between these forces is called:

Hydrostatic Equilibrium

If the Sun cools:

Pressure decreases.

The Sun contracts.

If fusion increases:

Pressure increases.

The Sun expands.

The pressure inside the Sun can be calculated using gravitational equilibrium.

The central pressure of the Sun is approximately:

3×10^9 atmospheres

Helioseismology

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Small disturbances in the Sun produce vibrations called:

Sun-quakes

The study of these vibrations is called:

Helioseismology

Scientists study solar vibrations using Doppler shifts of hydrogen lines.

These vibrations help scientists understand the interior structure of the Sun.

Temperature of Planets

The temperature of planets depends on solar radiation.

The Stefan-Boltzmann law states:

Radiated power $\propto T^4$

where:

T = temperature

For Earth:

Solar temperature:

5800 K

Earth's calculated temperature:

280 K

Actual average surface temperature:

290 K

The difference is mainly due to:

Greenhouse Effect

Greenhouse Effect

The atmosphere reflects about:

30% of solar radiation

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Some gases trap outgoing heat radiation.

Important greenhouse gases:

Carbon dioxide (CO_2)

Methane (CH_4)

Water vapor (H_2O)

Nitrogen, oxygen, and argon do not strongly absorb solar radiation.

CO_2 concentration has increased due to industrial activities.

The greenhouse effect increases Earth's temperature.

Quantum Mechanics and Greenhouse Gases

Quantum mechanics explains why certain gases absorb radiation.

Molecules can exist only in specific energy states.

Molecules such as:

CO_2

H_2O

can absorb infrared radiation.

Their energy levels are quantized.

The absorbed radiation matches the infrared radiation emitted by Earth.

Therefore, greenhouse gases trap heat.

Quantum mechanics accurately predicts molecular absorption frequencies.

Future of the Sun

The Sun will eventually die because its hydrogen fuel will run out.

The Sun can continue hydrogen fusion for approximately:

5 billion years

During this period:

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Hydrogen converts into helium.

Helium accumulates in the core.

When hydrogen decreases:

Fusion rate decreases.

Gravity causes the core to shrink.

The core temperature increases.

Red Giant Stage

After hydrogen exhaustion:

A new fusion reaction begins around the core.

The Sun expands greatly.

It becomes a:

Red Giant Star

The diameter of the Sun increases from:

1/100 AU

to approximately:

1 AU

The Earth will be affected by this expansion.

Helium Fusion

Helium can fuse to produce carbon.

Helium fusion requires higher temperatures.

Carbon fusion begins at approximately:

600 million K

Low mass stars cannot reach this temperature.

The Sun will eventually lose its outer layers and end its life as a smaller star.

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

20 Difficult Statement, Formula & Numerical MCQs

1. The main source of energy of the Sun is:

- A) Chemical burning
- B) Nuclear fission
- C) Nuclear fusion ✓**
- D) Gravitational energy

2. The approximate age of the Sun is:

- A) 1 billion years
- B) 4.6 billion years ✓**
- C) 10 billion years
- D) 100 million years

3. The mass of the Sun is approximately:

- A) 2×10^{20} kg
- B) 2×10^{30} kg ✓**
- C) 2×10^{40} kg
- D) 2×10^{15} kg

4. The core temperature of the Sun is approximately:

- A) 5800 K
 - B) 100,000 K
 - C) 15 million K ✓**
 - D) 1 million K
-

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

5. The surface temperature of the Sun is:

A) 15 million K

B) 5770 K ✓

C) 100 million K

D) 300 K

6. The Sun's composition mainly contains:

A) Carbon and oxygen

B) Hydrogen and helium ✓

C) Iron and silicon

D) Nitrogen and oxygen

7. Solar energy is produced by converting mass according to:

A) $F = ma$

B) $E = mc^2$ ✓

C) $PV = nRT$

D) $\lambda = h/p$

8. The Sun converts hydrogen into:

A) Oxygen

B) Carbon

C) Helium ✓

D) Nitrogen

9. Nuclear fusion occurs mainly in the:

A) Surface

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

B) Core ✓

C) Radiation zone

D) Atmosphere

10. The process of combining atomic nuclei is called:

A) Radiation

B) Decay

C) Fusion ✓

D) Ionization

11. The two forces balanced in the Sun are:

A) Magnetic and electric forces

B) Gravity and gas pressure ✓

C) Nuclear and chemical forces

D) Heat and radiation forces

12. The balance of forces inside the Sun is called:

A) Quantum state

B) Hydrostatic equilibrium ✓

C) Nuclear equilibrium

D) Thermal radiation

13. The study of vibrations of the Sun is called:

A) Cosmology

B) Astronomy

C) Helioseismology ✓

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D) Spectroscopy

14. Stefan-Boltzmann law states that radiation is proportional to:

- A) T
 - B) T^2
 - C) T^4 ✓
 - D) T^3
-

15. The main cause of greenhouse effect is:

- A) Nitrogen
 - B) CO_2 , CH_4 and H_2O gases ✓
 - C) Oxygen
 - D) Argon
-

16. Quantum mechanics explains greenhouse effect through:

- A) Nuclear reactions
 - B) Molecular energy levels ✓
 - C) Gravity
 - D) Magnetic fields
-

17. The Sun will become a Red Giant after:

- A) Increasing hydrogen supply
 - B) Exhaustion of core hydrogen ✓
 - C) Losing all helium
 - D) Cooling completely
-

PHY101 IMPORTANT NOTES + EXAM KEY POINTS + LECTURE WISE MCQS

18. Helium fusion requires temperature approximately:

- A) 5800 K
 - B) 15 million K
 - C) **100 million K** ✓
 - D) 300 K
-

19. The energy produced by nuclear fusion comes from:

- A) Increase in mass
 - B) **Mass defect** ✓
 - C) Chemical reactions
 - D) Friction
-

20. The Sun can continue hydrogen fusion for about:

- A) 1 million years
 - B) 500 million years
 - C) **5 billion years** ✓
 - D) 100 billion years
-

THE END

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